Environ Monit Assess (2026) 198:61
https://doi.org/10.1007/s10661-025-14912-8

RESEARCH

Land use and land cover scenarios and their interrelations with sediment production and transport—Doce River basin

Murilo Brazzali Rodrigues · Izabela Tavares Spagnol · Livia Luchi Rabello · Selena Lopes Carvalho · Mônica de Souza Mendes · Diogo Costa Buarque · Antônio Sergio Ferreira Mendonça · Daniel Rigo · Marco Aurélio Costa Caiado

Received: 12 August 2025 / Accepted: 11 December 2025
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025

Abstract Information on sediment generation and transport is essential for identifying areas vulnerable to erosion and for planning conservation interventions in river basins. This study focuses on the Doce River basin, in Southeastern Brazil, a region predominantly covered by pasturelands and historically affected by anthropogenic activities, even before the collapse of the Fundão dam in 2015. The objective of the research was to evaluate different land use and land cover scenarios and their impact on sediment generation and transport, using the MapBiomas dataset and the MGB-SED hydrosedimentological model. Four scenarios were modeled: (1) historical land use changes, (2) and (3) enforcement of Brazilian environmental legislation, and (4) implementation of conservation practices in pasture areas. The results from scenario 1 showed that land use changes over time were insignificant. Scenarios 2 and 3 showed reductions of 11% and 15%, respectively, in sediment generation and transport compared to scenario 1. Scenario 4 achieved the most significant impact, with a 32% reduction, considering less restrictive conservation practices (P MUSLE 0.75). These results highlight the importance of improving pasture management to mitigate sediment generation and transport. Furthermore, the study identified priority sub-basins, Caratinga, Piracicaba, Casca, Santo

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10661-025-14912-8.

M. B. Rodrigues · I. T. Spagnol · S. L. Carvalho · M. d. Mendes (✉) · D. C. Buarque · A. S. F. Mendonça · D. Rigo
Department of Environmental Engineering, Federal University of Espírito Santo (Ufes), Fernando Ferrari ave., 514, Vitória-ES 29075-910, Brazil
e-mail: monicasmcastro@gmail.com

M. B. Rodrigues
e-mail: murilobrazzali@gmail.com

I. T. Spagnol
e-mail: izabelaspagnol.engamb@gmail.com

S. L. Carvalho
e-mail: selenaesa@gmail.com

D. C. Buarque
e-mail: diogo.buarque@ufes.br

A. S. F. Mendonça
e-mail: anserfm@terra.com.br

D. Rigo
e-mail: daniel.rigo@ufes.br

L. L. Rabello
Sanitary and Environmental Engineering Coordination, Federal Institute of Education, Science, and Technology of Espírito Santo (Ifes), Vitória ave., 1729, Vitória-ES 29040-780, Brazil
e-mail: eng.liviarabello@gmail.com

M. A. C. Caiado
Coffee Technology Coordination, Federal Institute of Education, Science, and Technology of Espírito Santo (Ifes), Road ES-482, Km72, Alegre-ES 29500-000, Brazil
e-mail: mcaiado@ifes.edu.br

Published online: 21 December 2025

Antônio, and Carmo, as highly susceptible to erosion and major contributors to sediment production. These areas require restoration and environmental recovery actions, with an emphasis on soil conservation, to enhance environmental resilience and reduce sediment-related impacts throughout the Doce River basin.

Keywords Hydrosedimentological modeling · Soil erosion · Priority basins · MapBiomas · MGB-SED

Introduction

Sediment generation and transport are natural processes that, when intensified by human activities, can lead to environmental degradation and reduced soil productive capacity (Liu et al., 2023; Yin et al., 2021). Detailed information on these processes is essential for identifying areas vulnerable to erosion and for planning conservation interventions in river basins (Rautela, 2021; Admas et al., 2022; Dananto et al., 2022).

The Doce river basin already had a history of negative environmental impacts resulting from various anthropogenic activities (Rodrigues et al., 2013) and stands out for having one of the highest concentrations of suspended solids in Brazil, with an average of 386.3 mg·L^{-1} recorded at the Colatina station (ES). This value significantly exceeds the average observed in the Amazon River basin, which is 149.3 mg·L^{-1} at the Óbidos station (PA) (Lima et al., 2005). The magnitude of the impacts observed in the Doce River basin, evidenced by the significant increase in the concentration of suspended solids, highlights the importance of understanding hydrosedimentological processes.

Additionally, on November 5, 2015, occurred the collapse of the Fundão dam (Mariana, MG), which represented the largest socio-environmental disaster in Brazil in terms of the volume of waste released (Carmo et al., 2017). The event directly impacted the Doce River basin. During the collapse, of the 56.6 million m^3 of stored waste, an estimated 43.3 million were released into the environment, contaminating 668 km of watercourses and reaching the coastal-marine system adjacent to the mouth of the basin (Kütter et al., 2023). In response, a landmark institutional effort was launched, culminating in the "Novo Acordo do Rio Doce" (New Doce River Agreement), formally signed on October 25, 2024. This agreement establishes a recovery framework structured around 42 integrated programs, several of which prioritize environmental restoration throughout the basin. The disaster-linked material (DLM) compromised water quality, soils, vegetation, and aquatic and riverine habitats, including biota (FEST, 2024a), generating long-term chronic impacts. The most intense impacts occurred up to the Risoleta Neves HPP reservoir, where the mud reached marginal areas up to 200 m deep and thicknesses of up to 10 m, such as in the Bento Rodrigues district. In total, 1026.65 hectares of vegetation cover were destroyed, including stretches of Atlantic Forest. This impact increased the contribution of sediments to water bodies, intensifying their silting (Kütter et al., 2023; Carmo et al., 2017; Silva, Cunha & Viglio, 2016). According to Hatje et al. (2017), estimates indicate an increase of approximately 33,000 mg·L^{-1} in the total concentration of suspended solids after the dam collapse, with much of this material settling on the banks and riverbed.

Studies show that changes in land use and land cover influence hydrosedimentological processes in river basins (Costa et al., 2023; Cunha et al., 2022; Gómez-Delgado et al., 2011; Silva et al., 2023a, 2023b). Elements such as the implementation of reforestation and soil conservation programs can lead to a substantial reduction in erosion (Chasia et al., 2024).

Wang et al. (2024) developed a dynamic model of slope erosion that integrates the effects of vegetation on soil detachment and sediment transport. The authors analyzed the impact of forests, pastures, and agricultural lands, varying the coverage rate of each of these uses between 20% and 60%, obtaining as results a reduction in sediments of 66.75%–99.95%, 57.43%–99.63%, and 27.91%–88.63%, respectively, for forests, pastures, and agricultural lands.

In addition, basins with forest cover and controlled agricultural practices, together with soil conservation, present the lowest rates of sediment production (Anley & Minale, 2024; Chasia et al., 2024; Kitheka et al., 2022; Liu et al., 2021; Zhang et al., 2023). Conversely, the conversion of natural areas to intensive agriculture and livestock farming increases erosion processes and the silting of river channels (Bayer et al., 2020).

In this scenario, modeling sediment transport and production is an essential tool for understanding and managing river basins (Huang et al., 2024), allowing the assessment of the impact of different land use scenarios, climate change, and human interventions in the basin (de Waal et al., 2024). Furthermore, it is possible to identify vulnerable and priority areas for environmental management and recovery (Mrad et al., 2024; Sime & Abebe, 2022).

A systematic review carried out by Andualem et al. (2023) mentioned several models of erosion, production, and sediment transport, classified into three main categories: empirical, conceptual, and based on physical processes. Some of the models cited include empirical models (USLE, RUSLE, MUSLE, PESERA, SEDD, and SLEMSA), conceptual models (STREAM, ANSWERS, MMF, and AGNPS), and models based on physical processes (SWAT, WEPP, SHETRAN, LISEM, GLEAMS, AnnAGNPS, and SHESED).

Among the available models, the MGB-IPH, a model based on physical processes, stands out for its Brazilian origin and capacity for automatic recovery of soil and vegetation data via satellite (Brito Neto et al., 2021; Buarque, 2015; Fagundes et al., 2019, 2020; Föeger, 2019). This model has been applied in several basins in Brazil, ranging from flow forecasting (Pessoa et al., 2012), water quality simulation (Fan et al., 2013; Melo, 2014), and studies on the impact of climate change (Nóbrega et al., 2011; Paiva et al., 2011). Furthermore, the model has the capacity to represent the hydrological impacts of land use changes (Bayer & Collischonn, 2013), which makes it particularly useful for analyzing the effect of vegetation change in large basins, where land use changes are not uniform (Coe, 2000; Eckhardt et al., 2002; Kucharik et al., 2000; Neitsch et al., 2011; Ragab & Bromley, 2010).

In the context of the MGB-IPH, the Large Basin Sediment Model (MGB-SED), developed by Buarque (2015) and improved in subsequent studies (Fagundes et al., 2019, 2020; Föeger, 2019), was created to simulate erosion and sediment transport in large-scale river basins. Using a simplified approach, it allows estimating the spatial and temporal distribution of sediments in river basins, as well as their movement through rivers, including interactions with floodplains (Buarque, 2015).

The model has been widely applied in hydrosedimentological studies in river basins, showing favorable performance. Fagundes et al. (2021) used the MGB-SED model to evaluate the spatial and temporal dynamics of the natural transport of suspended sediments in South America. The model estimated that South America exports approximately 1.00×10^9 t·year^{-1} of suspended sediments to the oceans. In the study, MGB-SED showed superior performance compared to other global models and regional studies in representing sediment fluxes. Fagundes et al. (2020) estimated the temporal and spatial variation of sediment transport along the Doce River basin, considering erosion and deposition processes in the rivers, using MGB-SED. Föeger et al. (2022) found that MGB-SED was able to satisfactorily represent sediment transport when studying the hydrosedimentological dynamics of the Madeira River basin, the most important southern component of the Amazon basin.

Considering the history of degradation of the basin, the collapse of the Fundão dam, and the New Rio Doce Agreement, this study seeks to provide scientific support for the decision-making process; this study directly contributes to these recovery efforts by evaluating normative and management-driven land use scenarios, including compliance with the Brazilian Forest Code through the "Cadastro Ambiental Rural" (CAR, Rural Environmental Registry), and identifying priority sub-basins for intervention. Rather than modeling the disaster's direct sedimentary impacts, this work addresses the structural, long-term drivers of sediment production and offers actionable insights to guide restoration under the New Doce River Agreement.

Materials and methods

The methodology of this work involved obtaining and processing land use data from MapBiomas (MapBiomas Project, 2024), and declaratory geospatial data from Permanent Preservation Areas (APP) and Legal Reserve (RL) from CAR, which are important for complying with current Brazilian legislation, especially the Brazilian Forest Code (Brasil, 2012). Based on these data and information from the river basin, scenarios were proposed for simulating sediment generation and transport with the MGB-SED model

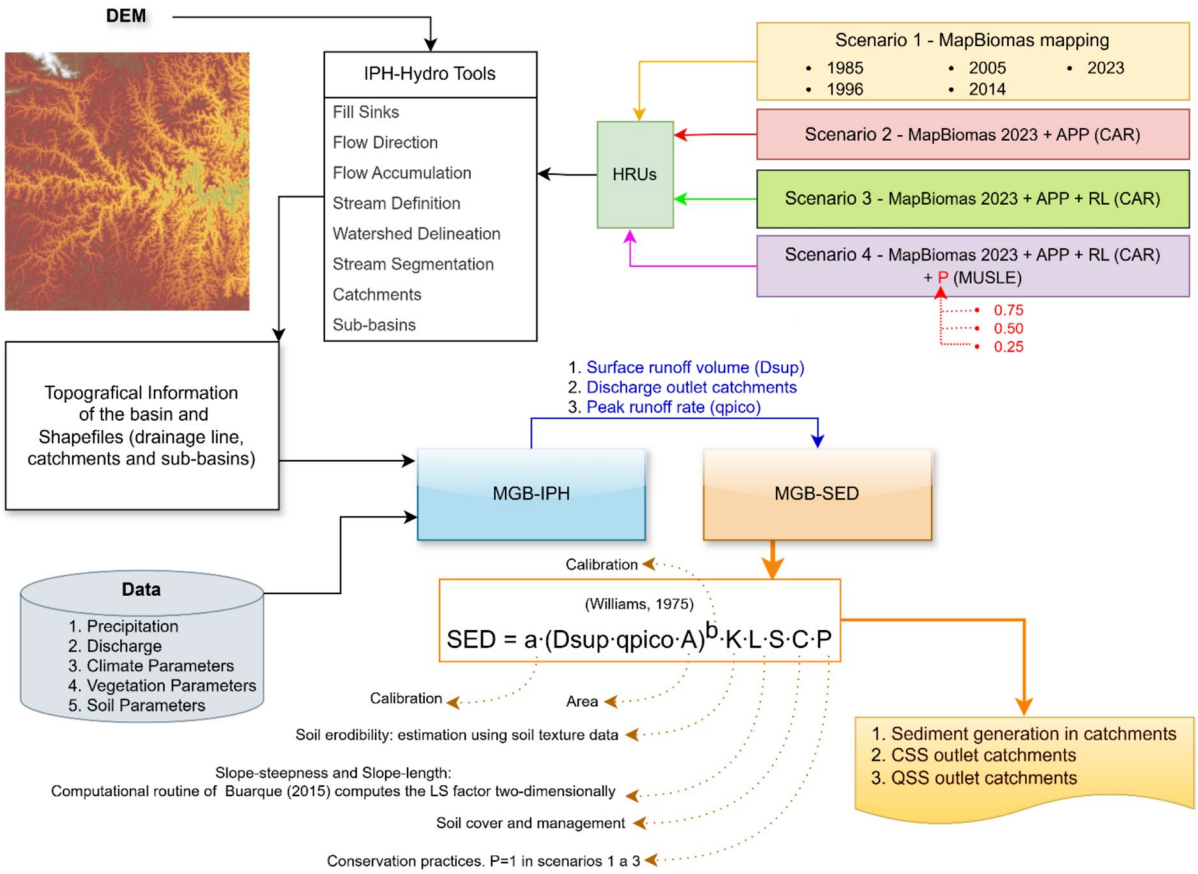


Fig. 1 Illustration of the methodological procedures adopted

(Buarque, 2015; Fagundes et al., 2019, 2020, 2021; Föeger et al., 2022). The procedures used in this work are illustrated in Fig. 1.

Study area: Doce River basin

The Doce River basin (Fig. 2), located in Brazil, has a drainage area of 83,400 km^2, of which 86% is in Minas Gerais and 14% in Espírito Santo. The Doce River is an interstate watercourse with an extension of approximately 875 km. Its sources are located in the Serra da Mantiqueira, in the municipality of Ressaquinha, in Minas Gerais, at an altitude of 1200 m above sea level, and its mouth is located in the municipality of Regência, in the state of Espírito Santo. The basin has a tropical highland climate, divided into three subtypes: cold summers in the highest areas, mild summers in intermediate altitudes, and hot summers in the regions of lower elevation (Felippe et al., 2016).

The upper region of the Doce River basin is the one with the highest volume of precipitation, where its forming rivers, Carmo and Piranga, and the Santo Antônio and Piracicaba rivers are found. The rainy season in the basin occurs from October to March, with the highest rainfall records in December and January (FEST, 2024b). The hydrographic basin studied presents variations in the values recorded for precipitation. The annual precipitation total ranges from 953 to 2450 mm, with an average value of approximately 1400 mm. The highest volumes of rainfall occur in the extreme southwest of the basin, with average annual precipitation exceeding 2000 mm, while in the northern and northeastern portions of the territory, the lowest volumes of rainfall are observed, with annual precipitation below 1200 mm (ENGECORPS ENGENHARIA, 2021).

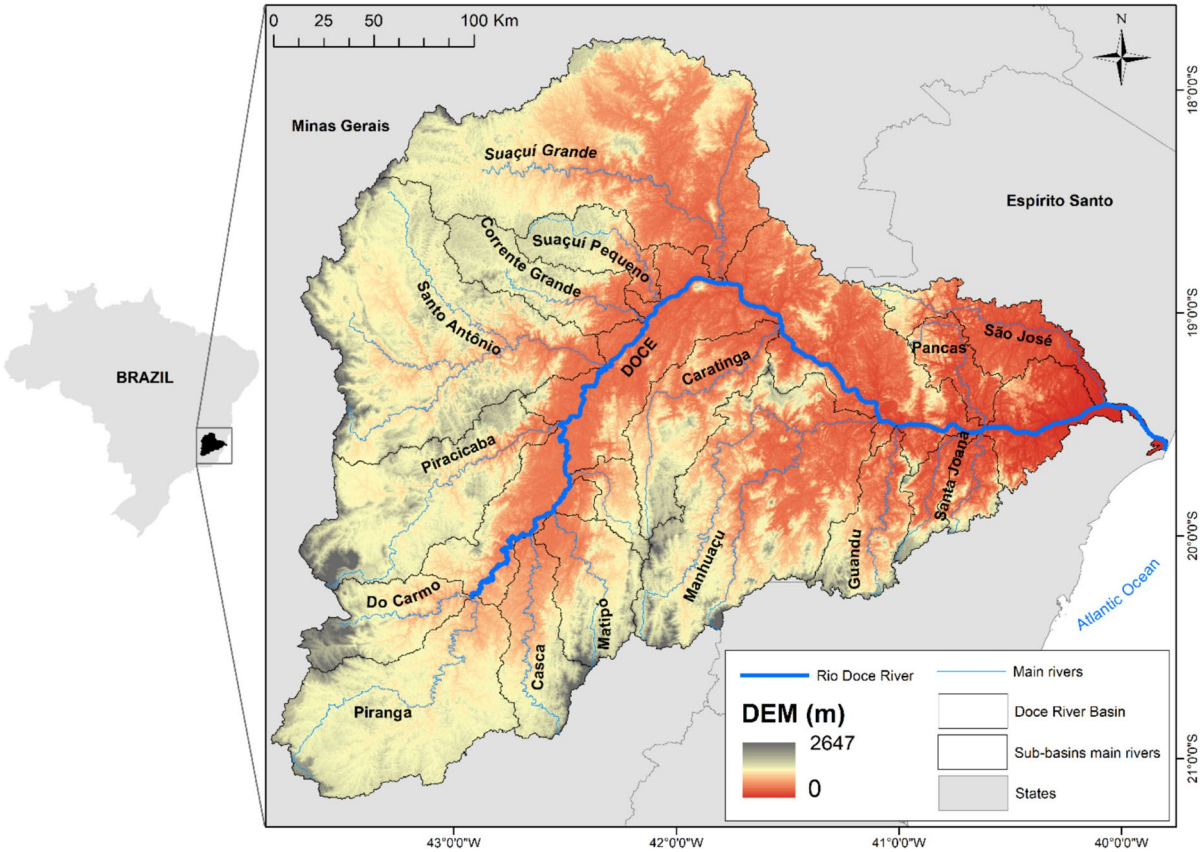


Fig. 2 Location of the portion of the Doce River basin located upstream of Colatina and identification of its main tributaries superimposed on elevation data

According to the Integrated Water Resources Plan for the Doce River Basin (PIRH Doce) (ENGECORPS ENGENHARIA, 2021), the Doce River basin has a complex distribution of erosive susceptibility, with 58.4% of the basin area classified as highly susceptible, influenced by heavy rainfall, sensitive soils, and morphodynamical processes such as ravines and gullies. Susceptibility varies, with 7.5% of the area classified as very highly susceptible, 29.6% as moderate, and only 0.9% as low.

The Doce River basin has undergone significant changes due to the historical process of continuous removal of the original native vegetation—the Semideciduous Seasonal Forest, which is the main plant formation of the Atlantic Forest biome. This process was intensified from the beginning of the twentieth century, with the construction of the Vitória–Minas Railway (EFVM). The construction of this railway enabled the effective occupation of the basin's territory, which was already suffering from deforestation, boosting the development of the regional economy. In this context, agricultural activities, iron ore extraction, the steel industry, and, from the 1940s onwards, the cultivation of eucalyptus for forestry purposes stand out.

Currently, agriculture occupies approximately 11% of the basin's area, concentrated in the regions of Piranga, Caratinga, and Manhuaçu (Minas Gerais portion) and in the regions of Pontões and Lagoas do Doce River and Barra Seca and Foz do Doce River (Espírito Santo portion) (Coelho, 2009; ENGECORPS ENGENHARIA, 2021).

Data collection

Satellite images of land use data for the years 1985, 1996, 2005, 2014, and 2023 were obtained through the MapBiomas project (MapBiomas Project, 2024),

which annually maps land cover and use in Brazil and makes them publicly, openly, and freely available. The time frame adopted in this work was intended to maintain an average interval of 8 years between images. The mappings for the years 1985 and 2023 were chosen because they are the borderline years with data available in the MapBiomas historical series. The 2014 mapping was defined as a reference, due to the Fundão dam collapse at the end of 2015.

MapBiomas data are validated with high-resolution images and fieldwork. Several authors used these data in their works, such as Ferreira Neto et al. (2024), Arruda et al. (2024), Souza Júnior et al. (2024), Rocha et al. (2024), and Baeza et al. (2022).

The geospatial data, made available in shapefile format, delimiting Permanent Preservation Areas (APP) and Legal Reserve areas (RL), were obtained through a public consultation with the National Rural Environmental Registry System (SICAR) (Brasil, 2024). The CAR is a public and mandatory registry for all rural properties, with the objective of recognizing the land use and coverage of each property. The CAR was created by Law No. 12,651/2012 (Brasil, 2012), which established general rules on the protection of vegetation, APP and RL. According to the Brazilian Forest Code, various areas are designated as APP, including those near water bodies and other legally protected locations. Still according to the legislation, RL areas (areas covered by native vegetation) must be between 20% and 80% of the size of the property, depending on its location.

In this study, 183,294 registrations were analyzed. Of this total, 0.4% were disregarded due to cancellation in the CAR, either by judicial or administrative decision, overlapping and/or duplication of areas, errors in the declaration of land use of the property, or at the request of the owner.

MGB-SED model

The MGB-SED model (Buarque, 2015; Fagundes et al., 2020; Föeger, 2019) works integrated with the Large Basin Model (MGB), which is a rainfall-runoff model for basins larger than 1000 km^2 (Collischonn & Tucci, 2001; Paiva et al., 2011; Pontes et al., 2017). The MGB-SED has three modules: basin, rivers, and plains. In the basin module, the MUSLE equation is used to estimate the load generated in each Hydrological Response Unit (HRU) of each unit-catchment (drainage area of each watercourse) according to Eq. 1 (Williams, 1975). However, the MGB-SED considers that not all sediment load generated in the unit-catchment reaches the drainage network at the same time, so a time delay according to Eq. 2.

$$SED_{i,j} = \alpha \bullet \left(Dsup_{i,j} \bullet qpico_{i,j} \bullet A_{i,j}\right)^{\beta} \bullet K_j \bullet C_j \bullet P_j \bullet LS_{i,j} \quad (1)$$

$$QS^t_{i,j} = \frac{1}{TKS_i} \bullet VSED^t_{i,j} \quad (2)$$

In Eq. 1, $SED_{i,j}$ is the sediment load resulting from soil erosion, α e β are adjustment coefficients, $Dsup_{i,j}$ [mm·ha^{-1}] is the surface runoff volume, qpico [m^3·s^{-1}] is the peak runoff rate, A [ha] is the surface area, K [0.013·t·m^2.h·m^{-3}·t^{-1}·cm^{-1}] is the soil erodibility factor, C [-] is the soil cover and management factor, P [-] is the conservation practices factor, and LS [-] is the topographic factor. In Eq. 2, $QS^t_{i,j}$ is the total solid discharge [t·s^{-1}], $VSED^t_{i,j}$ [t] is the sediment volume in the reservoir of the HRU j of the unit-catchment i in question, and TKS_i [s] is the delay time of the surface reservoir.

In the river's module, the transport of fine sediments (silt and clay) is carried out in a suspended manner, and the sand is transported as bedload subject to erosion and deposition processes depending on the transport capacity of the runoff. In the plain's module, the interaction between the rivers and their floodplains is considered, allowing the lateral exchange of fine sediments. Details of the MGB and MGB-SED models can be found in Buarque (2015); Fagundes et al. (2019, 2020, 2021) and Föeger et al. (2022).

To apply the MGB and MGB-SED models, topological and vector information obtained through the processing of digital elevation models (DEM) is required. In this sense, the discretization of the Doce River basin was performed through the manipulation of the DEM, using the IPH-Hydro Tools add-on (Siqueira et al., 2016). The DEM made available by the Copernicus DEM—Global and European Digital Elevation Model (COP-DEM) program (ESA, 2023)—was adopted. Among the versions of COP-DEM available, the one with the best spatial resolution (30 m) was chosen. The following procedures were applied to the DEM: sink and destroy, flow

direction, flow accumulation, stream definition, watershed delineation, stream segmentation by fixed length (10 km), and unit-catchments delineation, both available in the IPH-Hydro Tools add-on of the Quantum GIS 3.38 program.

Once the spatial characteristics of the Doce River basin were obtained, the HRUs were defined, which are areas that present similar hydrological behavior and where surface runoff, soil erosion, and other hydrological behaviors in the river basins are accounted for (Kumar & Mishra, 2015). The HRUs were constructed considering pedological information (IBGE, 2006) and land use and land cover (MapBiomas and CAR).

Considering the limitation of up to 12 possible combinations of soil type and land use and land cover in the definition of HRUs, the soil types identified in IBGE (2006) were reclassified into four classes according to their capacity to generate surface runoff: deep soils, shallow soils, semi-impermeable soils, and water, according to criteria defined by Fan et al. (2015). Likewise, of the 37 classes of land use and land cover differentiated by MapBiomas in Brazil, 23 classes were identified in the Doce River basin, and of these, were reclassified into 8 classes: forests, pasture, agriculture, semi-impermeable areas, silviculture, mining, water, and other uses. Of the 32 possible classes obtained with the overlap of the soil type and land use and land cover classes, the result of the overlap was reevaluated to obtain the 12 possible HRU classes. Details of the reclassifications can be found in the supplementary material.

The main spatial data extracted refers to the delimitation of the basin, segmentation of the drainage network (1611 segments of 10 km), definition of unit-catchments (1611 unit-catchments), and sub-basins (60 sub-basins to allow hydrological and sediment calibration) (Fig. 3).

The MGB-SED model also requires hydrological, climatic, vegetation, and sediment data for its application. Regarding hydrological data, daily rainfall data from 101 stations located in the Doce River basin and in areas close to its limits were considered; daily flow data from 61 stations, used to verify and adapt the hydrological model; and suspended sediment data from 31 stations, used to verify and adapt the sediment model. Sediment monitoring frequency varied substantially among the 31 stations in the Doce River Basin. Approximately 29% of the stations had an average of 2 samples per year, 68% had around 3 samples per year, and only 1 station (CEMIG) provided almost daily records, with an average of 119 samples per year. Detailed information on the monitoring period and the annual number of samples for each station is provided in the Supplementary Material. The rainfall, sediment, and fluviometric data were obtained from the Hydrological Information System—HidroWeb (ANA, 2024). Additionally, temperature, pressure, insolation, relative humidity, and wind speed data were obtained from 20 stations located in the Doce River basin and in areas close to its limits, belonging to the database of the National Institute of Meteorology (INMET). The spatial distribution of the stations used in the study is shown in Fig. 4.

The fixed vegetation parameters used in the model include albedo, leaf area index, vegetation height, and surface resistance. These parameters vary depending on the type of vegetation and seasonality, and the values established by Fagundes et al. (2019) for the Doce River basin were reproduced.

Regarding the MUSLE factors, the *LS* factor was determined in the pre-processing stage for each pixel of the DEM, with a specific computational routine (Buarque, 2015). The erodibility factor (K) was estimated using the EPIC model equation (Williams, 1995), based on the percentages of sand, silt, clay, and carbon for each soil type, as adopted by Fagundes et al. (2019). These soil texture data were obtained from the Food and Agriculture Organization of the United Nations (FAO, 1971) and are presented in the supplementary material; the soil cover and management factor (C) and conservation practices factor (P) were obtained based on the work of Fagundes et al. (2019, 2020) and Lopes et al. (2022).

Once all the MGB-SED input data were defined, the hydrological module was initially adjusted. Hydrological calibration is performed by modifying the physical properties of the soil (e.g., water storage). These parameters can be variable per HRU and/or per sub-basin, being carried out for the period from 1990 to 2014 and the validation from 1970 to 1989. The performance of the hydrological calibration was verified using the Nash–Sutcliffe efficiency (NSE), the Nash–Sutcliffe efficiency coefficient of the logarithms (NSELog), and the percent bias (PBIAS) of the hydrographs, and the statistical results were classified according to Moriasi et al. (2007).

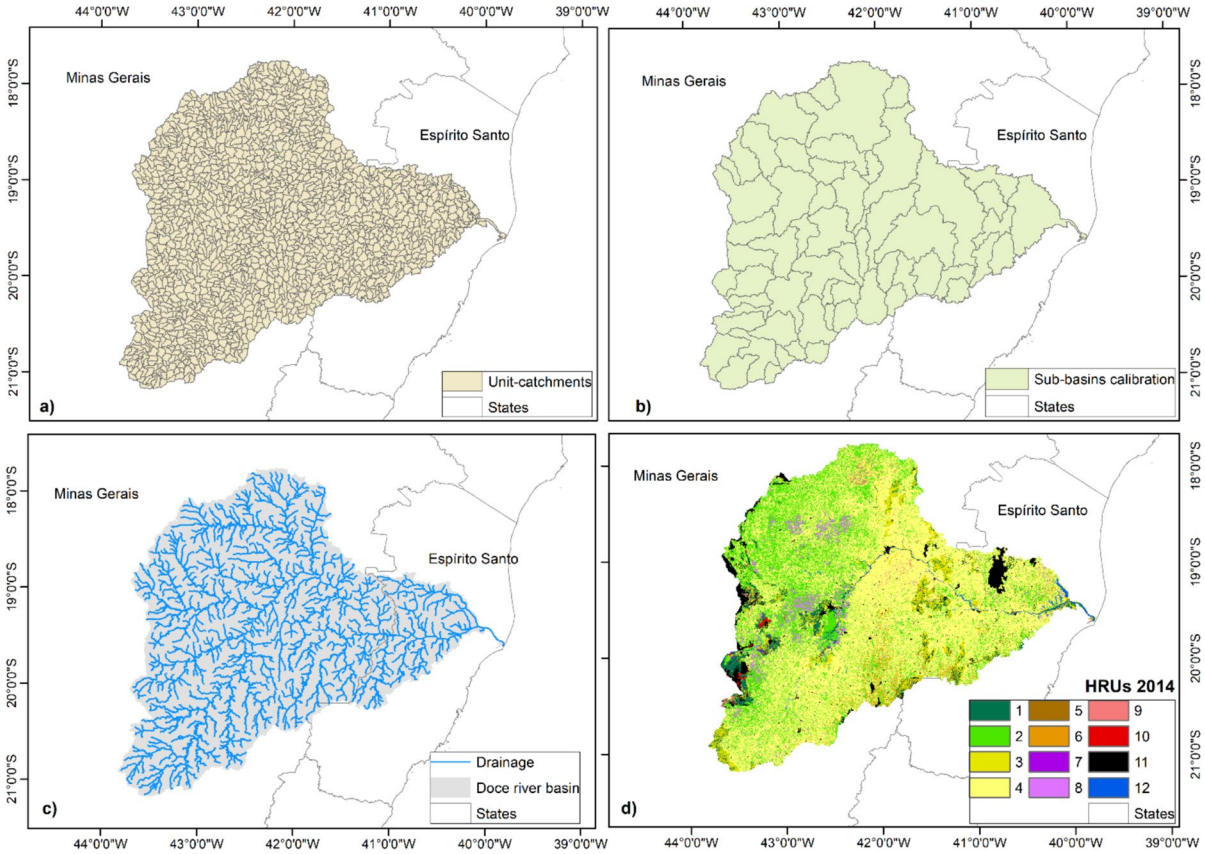


Fig. 3 Result of the discretization of the Doce River basin, with unit-catchment (**a**), calibration sub-basins (**b**), drainage sections (**c**), and HRU referring to the 2014 mapping (**d**). Notes (**d**): 1, forest formation and shallow soil; 2, forest formation and deep soil; 3, pasture and shallow soil; 4, pasture and deep soil; 5, agriculture and shallow soil; 6, agriculture and deep soil; 7, forest plantation and shallow soil; 8, forest plantation and deep soil; 9, other uses and deep soil; 10, mining; 11, semi-impervious area; 12, water

Calibration relative to sediments was performed from 1998 to 2014 (period in which most stations had available data). The parameters α, β and TKS_i were considered calibratable. The simulated and observed daily data of suspended sediment concentration (CSS $mg{\cdot}L^{-1}$) and suspended solids discharge (QSS $t{\cdot}d^{-1}$) were compared using sedimentograms and analyzed using statistical metrics such as the Kling-Gupta coefficient (KGE), coefficient of determination (R^2), and PBIAS, performance metrics already used by Aires et al., (2022, 2023) and Fagundes et al., (2019) in the Doce River. For sediment calibration, the MGB-SED model configuration was adopted, which uses the mapping of land use and land cover from 2014 (Fig. 1d), the year immediately before the collapse of the mining tailings dam, Fundão Dam. However, the calibration of the MGB-SED model was conducted with a primary focus on reproducing QSS, due to the greater uncertainties and high variability associated with CSS. Validation of the sediment model was not carried out due to the low availability of data, so it was decided to prioritize calibration, given that Fagundes et al. (2019) obtained lower results than the calibration period when validating the MGB-SED for the Doce River basin.

Simulated scenarios

A total of four simulation scenarios were defined to evaluate the dynamics of land use and land cover of the Doce River basin from 1985 to 2023,

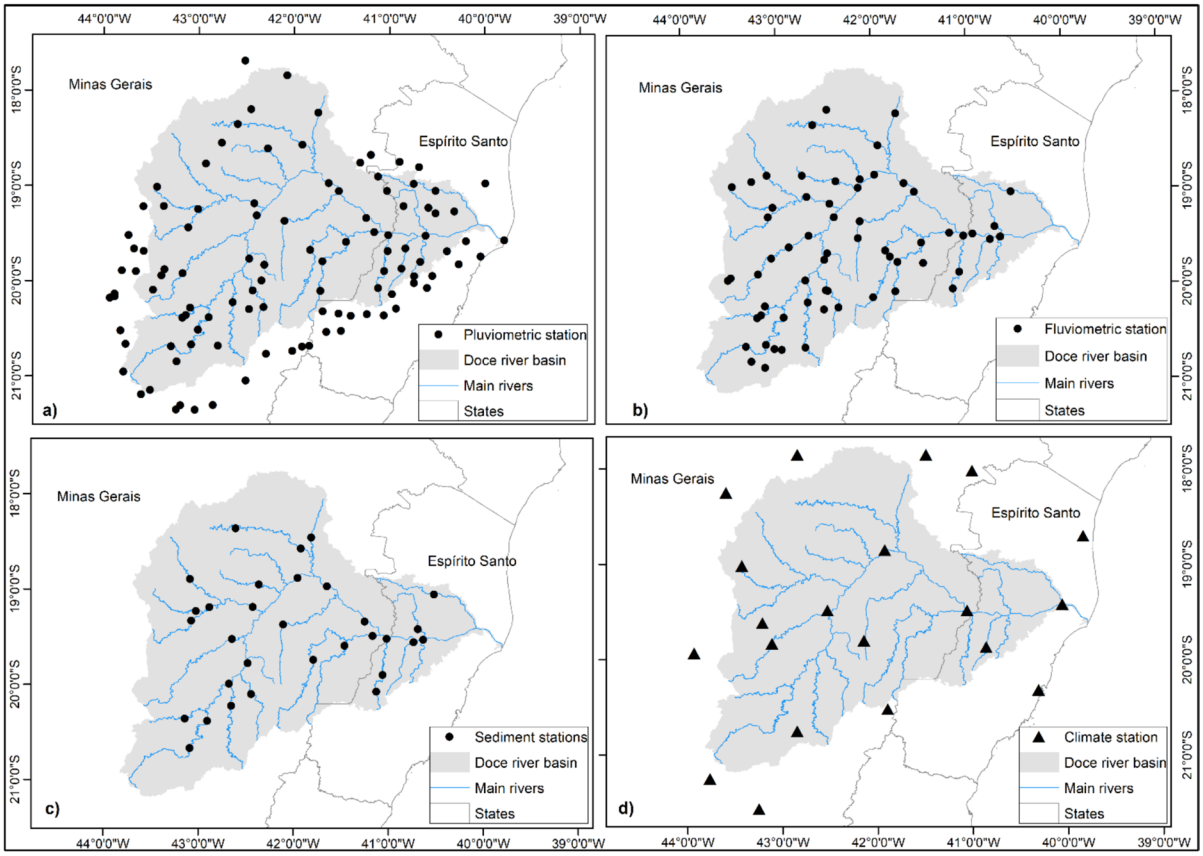


Fig. 4 Location of rainfall, fluviometric, sediment, and climate stations used in the hydrosedimentological simulation

including APP and RL areas and the adoption of conservation practices on pasture HRUs. Considering that the central part of the sediment modeling described in the previous section focuses on HRUs, all scenarios were implemented in the modeling by altering the HRU files, except for the conservation practices scenario, which focuses on a variable directly related to MUSLE. The simulation scenarios were as follows:

- Scenario 1: Simulations for each of the selected MapBiomas mappings (1985, 1996, 2005, 2014, and 2023), with the aim of verifying how changes in land use and cover affected the generation and transport of sediments over the years.
- Scenario 2: Considering the mapping of land use and coverage made available by MapBiomas for the year 2023, the APP areas (foreseen in the CAR, admitted as consolidated) were included as forest areas, with the aim of assessing whether the restoration of APPs, most of which are close to water bodies, would have a significant impact on reducing the generation and transport of sediments.
- Scenario 3: Considering the mapping of land use and coverage made available by MapBiomas for the year 2023, the APP and RL areas (foreseen in the CAR, admitted as consolidated) were included as forest areas.
- Scenario 4: Different simulations were performed on the HRUs developed in Scenario 3 (2023 + APP + RL), exclusively changing the values of the P factor (conservation practices) of MUSLE for HRUs exclusive to pastures, due to their high presence in the basin and their conservation condition being considered degraded. Values of 0.25, 0.5, and 0.75 were simulated. It is worth noting that the values were defined to evaluate a trend scenario, as there is no consistent data

for this factor for the basin; that is, in scenarios 1 to 3, all simulations adopted a *P* value of 1.

For scenario 4, the values were standardized for *P factor* in the MUSLE modeling for the Doce River basin due to the lack of detailed local data on conservation practices implemented in each sub-basin and the high spatial heterogeneity of land use and management conditions. Previous regional studies (Reda et al., 2024; Tsige et al., 2022; Watene et al., 2021, and Didoné et al., 2021) highlight that the P factor varies widely with slope, management type, and practice efficiency, and it is common to use literature-based representative values (e.g., 0.25, 0.50, 0.75) when site-specific field data are unavailable. Furthermore, reviews from Panagos et al. (2015) and Alexandridis et al. (2015) emphasize that parameter standardization is a methodologically acceptable approach for regional-scale erosion assessments, such as the Doce River basin, ensuring consistency and comparability across studies while focusing on prioritization rather than absolute soil-loss quantification. Therefore, even with generalized P values, the model provides relevant insights for identifying priority areas for vegetation restoration and soil conservation.

Additionally, to ensure that the changes implemented in the scenarios were the only changes allowed, all simulations were performed for the same period defined for calibration relative to sediments (1998–2014). After the completion of the hydrosedimentological calibration, no other changes were made to the model parameters.

Criteria for identification of priority areas

Priority areas for recovery were defined using a hierarchical set of criteria: (i) intensity of sediment generation and transport; (ii) predominance of pasture land use; (iii) average annual rainfall; and (iv) degree of erosion susceptibility. The erosion susceptibility classification was developed solely from vector data on soil, geomorphology, and precipitation (ECOPLAN-LUME, 2010), excluding land use and land cover. The integration of these factors enabled the identification of the most vulnerable sub-basins, guiding the prioritization of restoration actions.

Results and discussion

Evolution of land use and land cover in the Doce River basin

According to the maps made available by MapBiomas between 1985 and 2023, it was observed that there was no significant change in the evolution of land use and land cover in the basin, as shown in Fig. 5. Throughout the period analyzed, the basin was predominantly composed of pastures (approximately 60%), followed by forests (approximately 30%) and semi-impermeable areas (approximately 3.5%), which include rocky areas and urban areas. Although the results indicate an increase in agricultural areas (approximately 2%) and forest areas (approximately 3%), the percentage numbers are minimal when compared to those relating to the pasture and forest classes.

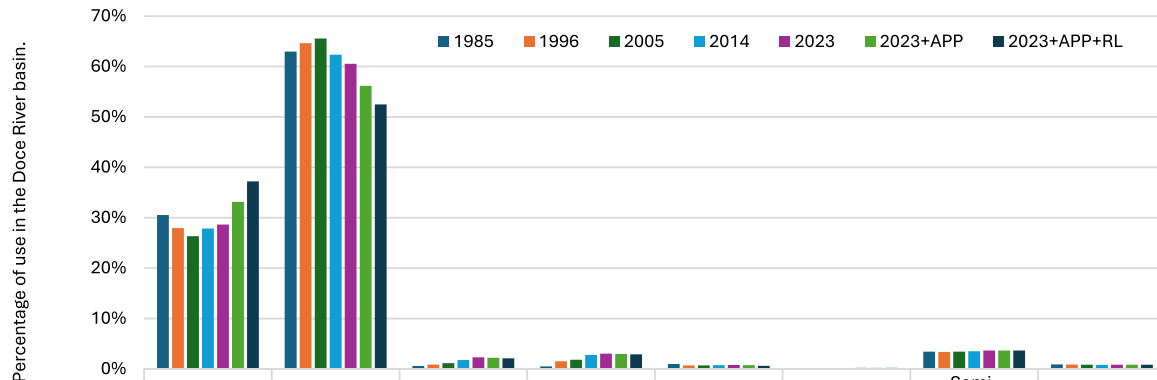
By 2023, with the inclusion of APP areas, there was a 4.5% increase in forest areas, to the detriment of pasture, agriculture, and forestry areas. With the addition of RL areas, there was a 4% increase in forest areas, compared to the scenario with APP only (scenario 2), with a 3.7% reduction in pasture areas. Thus, scenario 3, which represents the full implementation of the CAR APP and RL areas, would provide an 8.6% increase in forests, mostly in pasture areas, whose reduction was 8%.

The changes caused by the inclusion of APP and RL areas (Fig. 6) were concentrated along the banks of rivers and streams (APP), where recovery depends on the width of watercourses and around springs, while RL areas were concentrated together with existing forest remnants.

Estimates of sediment generation and transport

Hydrosedimentological calibration

The first stage of the modeling process involved the hydrological calibration of the model, which, according to the classification of Moriasi et al. (2007), at least 89% of the stations used for daily hydrological calibration presented NSE values considered at least "satisfactory" (NSE/NSElog > 0.5). In the validation, the percentage of stations was close to 86%; however, regardless of the evaluation, at least 30% of the stations presented NSE values considered "very good"



	Forest Formation	Pasture	Agriculture	Forest Plantation	Other uses	Mining	Semi-impervious area	Water
1985	30.5%	63.0%	0.6%	0.5%	1.0%	0.1%	3.4%	0.9%
1996	27.9%	64.6%	0.9%	1.5%	0.7%	0.1%	3.4%	0.9%
2005	26.3%	65.6%	1.2%	1.8%	0.7%	0.1%	3.4%	0.8%
2014	27.9%	62.3%	1.8%	2.8%	0.8%	0.2%	3.5%	0.8%
2023	28.6%	60.5%	2.3%	3.1%	0.8%	0.2%	3.6%	0.8%
2023+APP	33.2%	56.2%	2.2%	3.0%	0.7%	0.2%	3.6%	0.8%
2023+APP+RL	37.2%	52.5%	2.1%	2.9%	0.6%	0.2%	3.6%	0.8%

Fig. 5 Percentage evolution of land use and land cover across selected MapBiomas mappings and inclusion of APP and RL information from CAR

(NSE/NSElog > 0.75). Regarding NSELog, at least 90% and 77% of the stations presented statistical values considered "satisfactory" in the calibration and validation, respectively. The smallest errors related to volumes (PBIAS) were found in the calibration, with 98% of the stations presenting values considered "satisfactory" and 64% considered "very good." In the validation, the PBIAS values considered "satisfactory" (PBIAS < ±25) reduced to 88% of the stations and 29% of the stations considered as "very good." In this sense, the analysis of the results of the performance metrics, hydrographs, and permanence curves indicated that the model satisfactorily represents the hydrological dynamics of the basin under study. The hydrograph and permanence curve for the Colatina fluviometric station (the closest to the Doce River mouth considered in this study) are presented in supplementary material, illustrating the good fit found with the model.

After the hydrological calibration was completed, the MUSLE parameters α, β and the retardation coefficient TKS_i were adjusted for calibration relative to sediments. KGE and R^2 values above 0.5 were obtained in 45% and 42% of the stations, respectively, when evaluating the suspended sediment concentrations (CSS). When considering the suspended solid discharge (QSS), more than 70% of the stations presented KGE and R^2 values above 0.5, considered in this work as "satisfactory" since the simulation scale was daily. This threshold is consistent with commonly adopted criteria for satisfactory model performance in daily simulations (Knoben et al., 2019; Moriasi et al., 2007, 2015; Towner et al., 2019). Regarding PBIAS, at least 71% of the stations presented values considered as "good" (PBIAS < ±30), when compared to suspended sediment concentration data, and 100% when compared to suspended solids discharge. To illustrate the adjustment of the MGB-SED model, the CSS and QSS sedimentograms for the Colatina station are presented in supplementary material.

Sediment generation

Table 1 presents estimates of specific sediment generation for the Doce River basin and its main

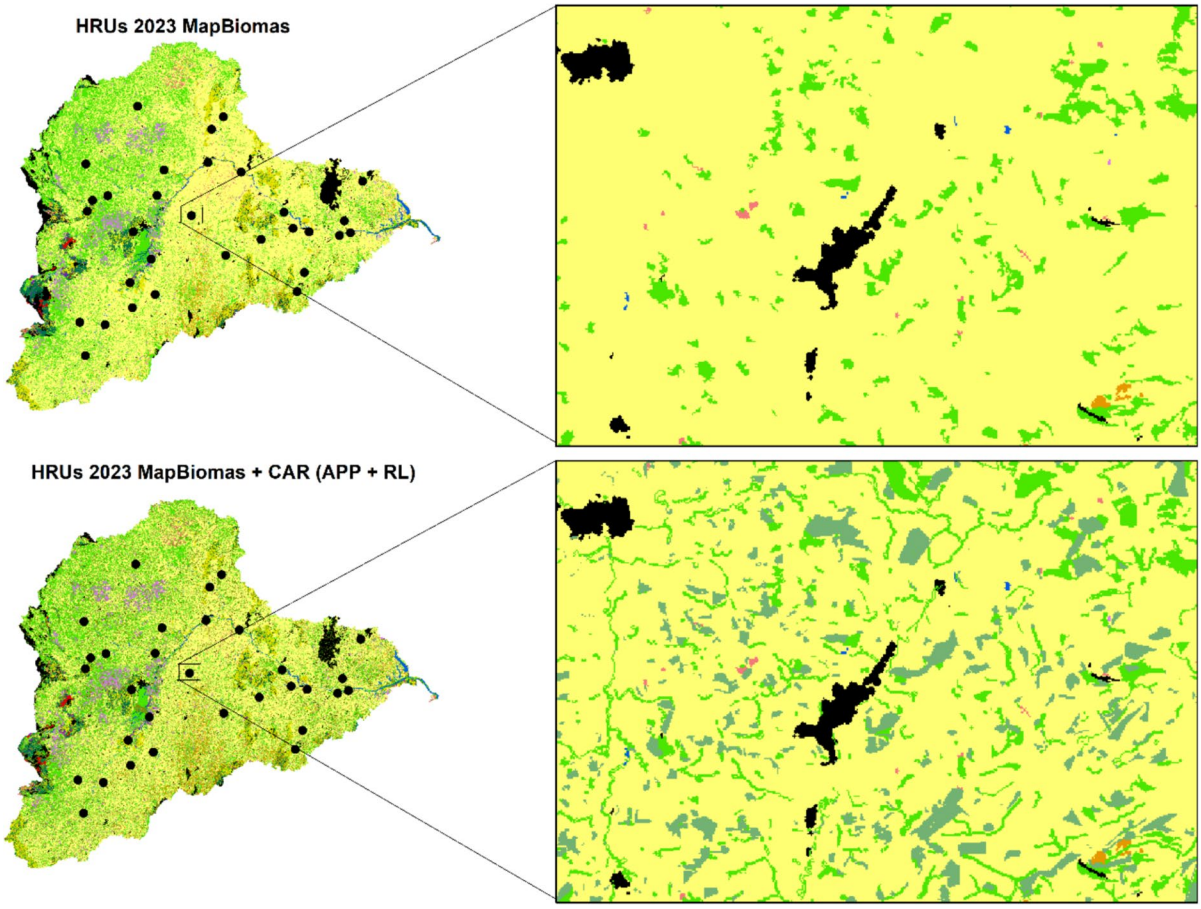


Fig. 6 Comparison of the 2023 maps with and without the inclusion of APP and RL areas declared in the CAR

sub-basins, considering the simulation scenarios. In scenario 1, considering only the initial and final years, the sub-basins of the Carmo, Piracicaba, Corrente, Caratinga, and Manhuaçu rivers showed an increase in specific sediment generation. In these basins, there was a reduction in forest and pasture areas (between 1% and 5%) and an increase of between 1% and 4% in agricultural and reforestation areas (characterized mainly by eucalyptus monoculture). Additionally, an increase of more than 70% in mining areas was recorded for the year 2023 compared to 1985.

In scenarios 2 and 3, with additions of APP and RL, Carmo, Casca, Piracicaba, Santo Antônio, Corrente, Caratinga, Guandu, and Manhuaçu sub-basins presented the highest estimates of specific sediment generation, with values ranging from 150 $t \cdot year^{-1} \cdot km^{-2}$ to 446 $t \cdot year^{-1} \cdot km^{-2}$. These scenarios, as previously mentioned, showed an increase in the forest class and a reduction in the pasture class, which promoted a reduction in sediment generation in relation to scenario 1, for the year 2023. Among the highest estimates, the Caratinga River sub-basin presented the greatest reduction in sediment generation, whose value was 82.8 $t \cdot year^{-1} \cdot km^{-2}$. Although the forest increases due to the incorporation of APPs and RL in the areas of these basins was 6% to 10%, they are located in regions presenting high rainfall rates, mining areas, mountainous and steep terrain (slope greater than 45%), and an average presence of more than 50% of pastures.

In scenario 4, which considers conservationist practices, a significant reduction in specific sediment generation was observed. As in scenario 3, the Caratinga River sub-basin presented the greatest reduction, 79 $t \cdot year^{-1} \cdot km^{-2}$, for coefficient $P = 0.75$, compared to scenario 3.

Table 1 Specific sediment generation (t·year^{-1}·km^{-2}) by sub-basin and basin of the Doce River by year of mapping and simulation scenario

Sub-basin (ID)	DrainageArea (km^2)	Scenario 1					Scenario 2	Scenario 3	Scenario 4		
		1985	1996	2005	2014	2023	2023+APP	2023+APP +RL	P=0.75	P=0.5	P=0.25
Piranga (1)	6639.3	83.9	84.0	84.2	82.1	79.7	68.8	66.0	51.8	37.6	23.4
do Carmo (2)	2296.2	217.7	223.9	228.2	228.1	223.8	192.8	185.8	157.2	128.5	99.9
Casca (3)	2526.2	253.7	258.2	258.6	258.2	252.7	212.0	201.1	157.2	113.4	69.5
Matipó (4)	2567.0	78.4	81.4	81.2	80.0	78.4	65.9	57.9	45.0	32.1	19.2
Sacramento (5)	841.8	85.6	85.9	85.8	84.2	84.1	77.3	72.2	56.3	40.4	24.6
Piracicaba (6)	5486.7	293.7	317.2	340.1	349.6	354.8	329.4	317.4	274.5	231.6	188.6
Santo Antônio (7)	10490.8	229.8	235.7	240.8	234.6	226.8	200.7	194.5	155.8	117.2	78.5
Corrente (8)	2494.7	153.4	153.1	170.8	154.1	155.7	146.9	142.8	114.9	86.9	59.0
Suaçuí Pequeno (9)	1732.4	94.8	92.5	97.5	89.0	87.0	78.5	76.1	61.2	46.3	31.4
Suaçuí Grande (10)	12479.9	125.2	128.9	131.8	123.7	114.7	103.9	98.3	77.1	55.9	34.7
Caratinga (11)	3246.7	437.6	449.4	454.0	453.6	446.5	385.6	363.7	284.7	205.6	126.6
Manhuaçu (12)	8888.1	168.3	173.4	175.3	177.4	170.6	150.8	143.3	113.7	84.2	54.6
Guandu (13)	2156.4	186.9	181.7	186.4	183.0	178.8	155.7	148.3	119.7	91.0	62.3
Santa Joana (14)	894.5	53.5	52.1	53.1	52.6	50.5	46.3	44.5	34.9	25.3	15.7
Santa Maria do Doce (15)	942.7	36.8	35.9	36.9	35.8	35.0	31.6	30.2	23.7	17.2	10.7
Pancas (16)	1186.3	24.0	24.3	24.2	23.0	22.9	21.5	20.2	15.9	11.6	7.3
São José (17)	2435.3	34.2	35.2	35.0	33.9	33.4	30.2	28.6	22.4	16.2	10.0
Other tributaries * (18)	15424.3	58.9	61.4	61.4	61.4	59.0	50.3	47.6	37.1	26.5	15.9
Doce River Basin	82729.4	151.4	155.9	159.9	157.6	153.5	136.0	129.9	104.6	79.2	53.9

*refers to the sum of drainage areas of several small watercourses that flow directly into the Doce River

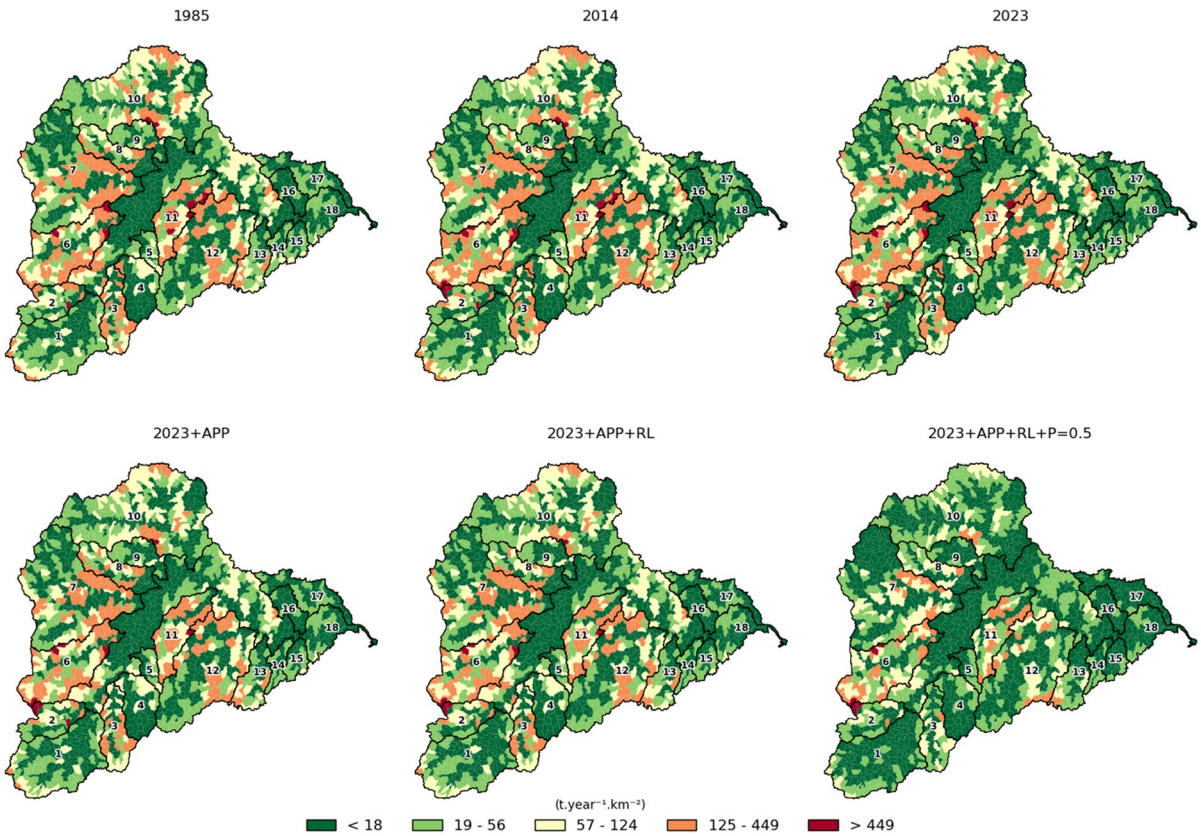


Fig. 7 Spatial representation of sediment generation for the Doce River basin and different simulation scenarios

Figure 7 shows the spatial distribution of sediment generation for the Doce River basin, identifying the sub-basins of its main tributaries. Sediment generation between the MapBiomas mappings presented in Fig. 7 for scenario 1 (1985, 2014, and 2023) indicated specific changes over the years, but insignificantly at the basin or sub-basins level. The visualization of sediment generation for the other years of scenario 1 is available in supplementary material (Fig. S3).

On the other hand, when incorporating the APP areas on the 2023 mapping, in scenario 2, a gradual reduction of 11% in sediment generation in the basin was observed, with some sub-basins showing percentages of 16% reduction (Matipó, Casca) compared to Scenario 1 (2023). In scenario 3, which incorporated the RL areas over scenario 2, the reduction in sediment generation was 15% over the load generated in 2023, with the sub-basins of the Matipó, Casca, and Caratinga rivers leading the reduction in sediment generation due to the APP and RL. These results reinforce the need to implement the APP and RL areas provided for in the CAR to improve the physical characteristics of the basin.

However, recognizing the importance of APP and RL areas, sediment generation in the basin is still high, since, as mentioned in the previous section, at least 50% of the basin is covered by mostly degraded pastures. In this context, combined with the lack of data on conservation practices in the study area, values were defined for the MUSLE P factor that makes up scenario 4. In this scenario, the results indicated that sediment generation in the Doce River basin is directly related to pasture areas, especially when evaluating the results for the value of $P = 0.75$, which resulted in a basin-level reduction in sediment generation of 49 $t \cdot year^{-1} \cdot km^{-2}$ when compared to scenario 1 (2023) and of 25 $t \cdot year^{-1} \cdot km^{-2}$ when considering the increases in APP and RL (scenario 3). A simple assessment of the results of scenario 4 in comparison to the others indicates that reducing sediment

generation in the Doce River basin requires actions aimed at improving pasture conditions.

Similar results were found by Qin et al. (2025), who observed a significant decrease in sediment generation in the Dali River basin, Loess Plateau, in China, and attributed this reduction mainly to changes in land use and land cover. In addition, the authors highlight that the vegetation restoration program, with an increase in forest and pasture cover, was identified as a crucial factor in reducing soil erosion. On the other hand, unlike these authors' conclusions, in the context of the Doce River basin, which has extensive areas of pasture, it is necessary to improve the conditions of this type of use to recover its soil protection capacity and exercise its erosion control function.

Correspondingly, Wei et al. (2022) identified the increase in vegetation cover as a determining factor in the reduction of sediment generation in the Upper Huaihe River basin, located in China, with a greater impact in areas of forest expansion. In addition, modeling studies, such as that of Jiang et al. (2024), have shown that the conversion of different types of land use can have varying effects on sediment generation. The conversion of steeply sloping agricultural land to forest has resulted in reduced sediment yield, and the conversion of grassland to shrubland and forest has also demonstrated potential to decrease sediment transport.

Sediment transport

Regarding sediment transport, the highest values of specific solid discharge were recorded in the regions that presented the highest sediment generation loads. Table 2 shows the specific suspended solids discharge ($t \cdot year^{-1} \cdot km^{-2}$) for the Doce River basin and main tributaries, by simulation scenario, and Fig. 8 shows the spatial distribution of sediment transport. In the analysis between the scenarios, it was observed that there was no significant difference between the different MapBiomas mappings. There was an improvement in the reduction of the sediment load transported by the basin to the Atlantic Ocean in scenarios 2 and 3, although limited to 11% with APP (scenario 2) and 15% with APP and RL (scenario 3), respectively. In scenario 4, with the inclusion of conservation practices on pastures, the reductions in sediment transport were significant when compared to scenario 1 (2023). In percentage terms, the reductions were 32%, 49%, and 65%, for P values equal to 0.75, 0.5, and 0.25, respectively.

Liu et al. (2024) found that soil and water conservation measures implemented in the Yellow River basin, in China, improved vegetation cover and reduced the sensitivity of sediment production to extreme precipitation. Vegetation has a relatively constant inhibitory effect on erosion over the years, while the distribution of precipitation varies significantly. On the other hand, in the Upper Blue Nile basin in Ethiopia, Anley and Minale (2024) reported a 68.9% increase in sediment transport, attributed to degradation of native vegetation, cultivation on steep slopes, and the lack of soil conservation practices. Similarly, in the Doce River basin, high sediment transport was observed in places with higher precipitation and areas with anthropized soils and steep slopes. Similarly, in this work, for scenarios 2, 3, and 4, in which there was an increase in forest areas and conservation practices were added to pastures, a reduction in sediment transport was observed.

Terassi et al. (2020) showed that sediment yield was more strongly correlated with event maximum discharge (Q_{max}) than with total precipitation and that seasonal changes in land cover exerted a greater influence on sediment mobilization than rainfall amount alone. Similarly, in rural basins of southern Brazil, Menezes et al. (2021) found that the relationship between precipitation, runoff, and suspended sediment concentration (SSC) is mediated by soil and land-use/management variables, highlighting how conservation practices and spatiotemporal changes in land use and vegetation cover modulated this relationship behavior.

The results align with field and modeling studies showing that improving degraded pasture conditions (rotational grazing, cover crops, or agroforestry) considerably reduces soil detachment and sediment transport (Didoné et al., 2021; Reda et al., 2024). In many Brazilian basins, degraded pastures are recognized as the main sediment source, and their rehabilitation offers greater sediment control than forest restoration alone (Oliveira et al., 2021). Also, these findings reinforce that structural and vegetative soil conservation practices are indispensable complements to reforestation efforts.

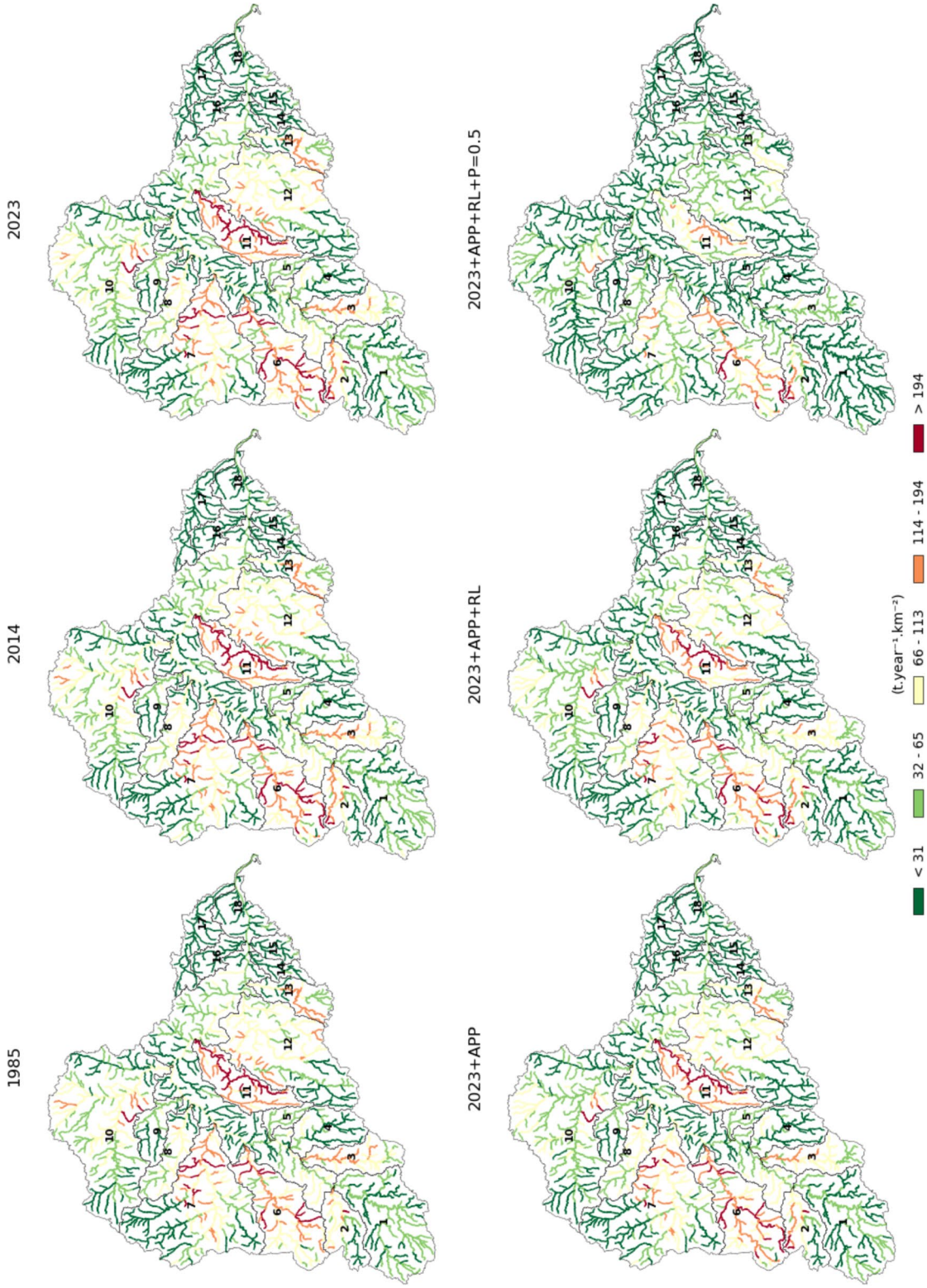


Fig. 8 Spatial representation of sediment transport in the Doce River basin for different simulation scenarios

Identification of priority areas

The spatial patterns for data on sediment generation and transport, land use and land cover, average annual rainfall, and susceptibility to erosion are illustrated in Fig. 9, which presents the 2023 mapping.

In this context, Table 3 presents the percentages of pastures and forests by sub-basin, since these uses together represent the majority of uses. Additionally, information on average annual precipitation (considering the same sediment simulation period) and classification regarding erosive susceptibility were also included. It is worth noting that erosive susceptibility classification was prepared using only vector information on soil, geomorphology, and precipitation; that is, data on land use and land cover were not incorporated (ECOPLAN-LUME, 2010).

In this study, approximately 83% of the sub-basins were classified as "highly susceptible" to erosion, with soils subject to erosion by ravines and gullies, either due to vegetation removal, steep slopes, or rainfall, according to the classification of the Integrated Water Resources Plan for the Doce River basin—PIRH (ECOPLAN-LUME, 2010). Areas of medium erosive susceptibility, mostly observed in the sub-basins of the Piranga, Carmo, and Corrente rivers, indicated that in these areas, the soils are subject to laminar, furrow, and gully erosion processes.

The sub-basins of the Caratinga, Piracicaba, Casca, Santo Antônio, and Carmo rivers presented the highest generation and transport of sediments, with pastures being the predominant use, most of them considered highly susceptible to erosion. Piracicaba, Casca, Santo Antônio, and Carmo are among the sub-basins presenting the highest rainfalls, which, in addition to representing the main erosive agent, find the soil unprotected due to poor soil conservation conditions. Carmo and Piracicaba also concentrate mining areas.

According to erosion susceptibility data presented by the PIRH (ECOPLAN-LUME, 2010), the Caratinga sub-basin, which is the largest in sediment generation and transport, has soils that are highly susceptible to erosion, which results in high sediment production, leading to landslides in locations with steeper slopes. The area of the Suaçuí Grande River sub-basin was classified as presenting "very strong" (31%) and "strong" (40%) susceptibility, being the one that presented the most areas classified as presenting "very strong" susceptibility. However, this classification did not coincide with the results of sediment generation and transport obtained in this work, presented in Tables 1 and 2. Although the Suaçuí Grande River is one of the main contributors of sediment to the Doce River, due to its extensive drainage area, its location coincides with the regions with the lowest average rainfalls in the basin, which justifies low sediment generation and transport compared to those corresponding to the other sub-basins.

In this context, the sub-basins of the Caratinga, Piracicaba, Casca, Santo Antônio, and Carmo rivers were the most vulnerable in terms of sediment generation and transport, emerging as the highest-priority targets for environmental recovery.

At the same time, the extensive area of the Suaçuí Grande basin, classified as "strongly susceptible" or "very strongly susceptible," also confers a certain degree of vulnerability. These areas with greater vulnerability are located among the highest in the basin, and due to their locations, restoration actions are part of the requirements of the Brazilian Forest Code, such as the restoration of APP and RL areas and, mainly, improvements in soil conservation, especially for the pasture class, as observed in the results relative to conservation practices.

Recent studies (Mendonça et al., 2025 and Reda et al., 2024) reinforce that under climate change effects, some Brazilian biomes are expected to maintain very strong levels of rainfall erosivity, with only minor variations over time. This indicates that in basins characterized by steep slopes and intense rainfall, the effectiveness of land-use or restoration measures may be constrained by persistently high erosivity. Consequently, sediment yield reduction scenarios should be interpreted within a context of enduring erosivity risk, highlighting the need for land-management strategies resilient to future increases in rainfall intensity and energy.

Overall, the results highlight that improvements in degraded pasture management are more effective in reducing sediment yield than passive forest recovery alone; the restoration of APP and RL areas offers essential local sediment control, and erosion hotspots coincide with steep, high-rainfall areas where combined measures are necessary. These insights are consistent with recent evidence emphasizing that integrated management of land use and climate risks is essential for effective sediment mitigation (Reda et al., 2024; Tsige et al., 2022; Watene et al., 2021).

Table 2 Specific sediment transport (t·year^{-1}·km^{-2}) by sub-basin and basin of the Doce River by year of mapping and simulation scenario

Sub-basin (ID)	DrainageArea (km^2)	Scenario 1					Scenario 2	Scenario 3	Scenario 4		
		1985	1996	2005	2014	2023	2023+APP	2023+APP +RL	P=0.75	P=0.5	P=0.5
Piranga (1)	6639.3	43.1	43.2	43.2	42.1	40.9	35.2	33.8	26.5	19.2	12.0
do Carmo (2)	2296.2	43.2	118.0	120.1	120.0	117.6	100.9	97.3	82.4	67.4	52.5
Casca (3)	2526.2	42.1	138.0	137.9	137.8	135.3	113.7	108.1	84.5	60.9	37.3
Matipó (4)	2567.0	40.9	40.7	40.6	40.2	39.5	33.2	29.9	23.2	16.5	9.9
Sacramento (5)	841.8	47.1	47.3	47.2	46.3	46.3	42.6	39.8	31.0	22.3	13.5
Piracicaba (6)	5486.7	144.7	155.1	166.8	171.1	174.0	161.3	155.5	134.6	113.7	92.8
Santo Antônio (7)	10490.8	118.6	121.4	124.1	121.1	116.9	103.4	100.2	80.3	60.4	40.5
Corrente (8)	2494.7	80.7	80.3	89.9	81.0	81.9	77.2	75.1	60.4	45.8	31.1
Suaçuí Pequeno (9)	1732.4	51.8	50.5	53.2	48.6	47.5	42.9	41.6	33.5	25.4	17.2
Suaçuí Grande (10)	12479.9	66.5	68.3	69.7	65.4	60.7	55.1	52.3	41.0	29.7	18.4
Caratinga (11)	3246.7	228.1	234.1	236.6	236.5	233.0	200.7	189.3	148.1	106.8	65.5
Manhuaçu (12)	8888.1	82.5	84.9	85.9	86.8	83.5	73.6	69.9	55.5	41.1	26.7
Guandu (13)	2156.4	91.8	89.5	91.8	90.3	88.2	76.7	73.2	58.9	44.6	30.3
Santa Joana (14)	894.5	27.9	27.2	27.7	27.6	26.5	24.3	23.4	18.3	13.2	8.2
Santa Maria do Doce (15)	942.7	19.2	18.7	19.3	18.7	18.3	16.6	15.8	12.4	9.0	5.6
Pancas (16)	1186.3	11.4	11.6	11.5	11.0	10.9	10.2	9.6	7.5	5.5	3.5
São José (17)	2435.3	13.8	14.2	14.2	13.7	13.5	12.2	11.6	9.1	6.6	4.1
Doce River Basin	82729.4	55.4	56.9	58.3	57.4	55.9	49.5	47.3	38.0	28.7	19.4

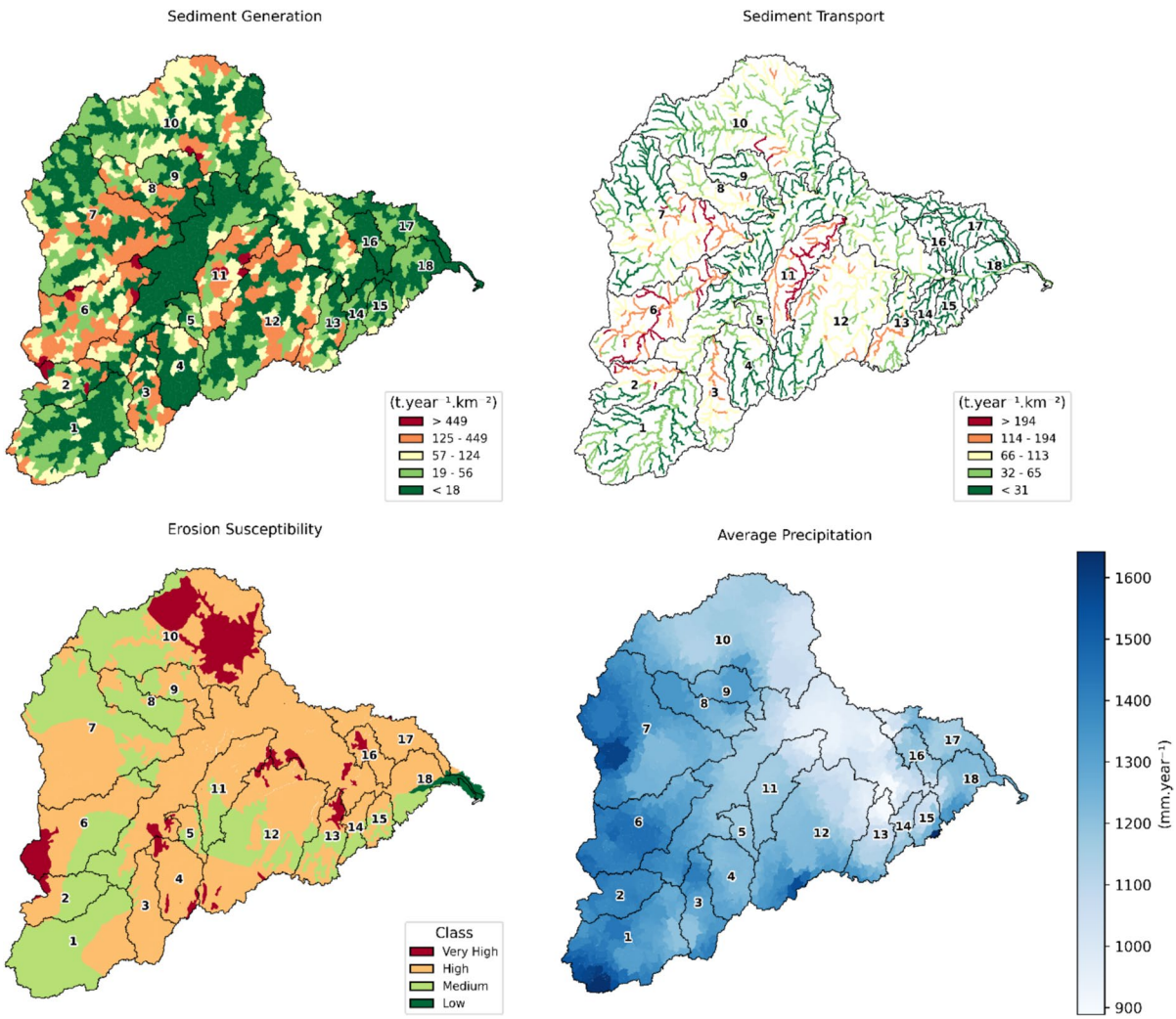


Fig. 9 Spatial representation of sediment generation and transport for the 2023 mapping, classification of erosion susceptibility, and average annual rainfall

Conclusion

The study analyzed the impacts of changes in land use and land cover on the generation and transport of sediments in the Doce River basin. It also investigated the effects of converting APPs and RLs into forest use, in addition to the adoption of conservation practices for pasture use. The adopted model, MGB-SED, showed good performance in simulating the generation and transport of sediments in the Doce River basin.

The work focused on the period from 1985 to 2023 (scenario 1), identifying that the changes that occurred in the different uses were not very significant, with a continued predominance of pastures (approximately 60%) and forests (approximately 30%).

Scenario 2, which included the adoption of APP areas for the year 2023, resulted in a 4.5% increase in forest use and a reduction in pastures, agriculture, and forestry, promoting an average reduction of 11% in sediment generation.

In turn, scenario 3, which represented the full implementation of the CAR APP and RL areas, would provide an 8.6% increase in the forest use class, which was mostly pasture areas that suffered

Table 3 Statistical summary of the main land uses (MapBiomas), average precipitation and erosive susceptibility (ECOPLAN-LUME, 2010) for each sub-basin of the Doce River

Sub-basin (ID)	1985		2014		2023		Average annual precipitation (mm/year)	Erosive susceptibility
	Forest	Pasture	Forest	Pasture	Forest	Pasture		
Piranga (1)	25%	74%	26%	69%	28%	68%	1381.9	Medium
do Carmo (2)	28%	67%	26%	68%	27%	66%	1392.1	Medium
Casca (3)	34%	67%	31%	67%	32%	65%	1323.0	Strong
Matipó (4)	43%	46%	39%	47%	38%	47%	1249.4	Strong
Sacramento (5)	22%	55%	23%	51%	24%	50%	1188.7	Strong
Piracicaba (6)	29%	65%	27%	64%	28%	61%	1391.7	Strong
Santo Antônio (7)	32%	63%	29%	62%	31%	60%	1356.1	Strong
Corrente (8)	45%	49%	40%	49%	41%	48%	1283.5	Medium
Suaçuí Pequeno (9)	27%	54%	24%	52%	25%	51%	1274.7	Strong
Suaçuí Grande (10)	33%	62%	30%	62%	31%	60%	1147.9	Strong
Caratinga (11)	36%	61%	33%	61%	33%	60%	1128.6	Strong
Manhuaçu (12)	33%	57%	29%	58%	30%	56%	1198.0	Strong
Guandu (13)	30%	73%	27%	70%	28%	68%	1109.0	Strong
Santa Joana (14)	20%	58%	19%	58%	20%	56%	1118.0	Strong
Santa Maria do Doce (15)	35%	50%	32%	52%	33%	50%	1158.0	Strong
Pancas (16)	38%	49%	32%	50%	32%	50%	1162.5	Strong
São José (17)	24%	67%	23%	66%	24%	63%	1154.1	Strong
Other tributaries (18)	28%	66%	26%	65%	27%	63%	1117.4	Strong

a reduction of 8%, which led to a reduction in sediment generation in the Doce River basin of approximately 15.4%.

For all simulated scenarios, with the exception of 4, it was assumed that pasture use is in a degraded state. In the simulation of scenario 4, it was assumed that in addition to the APP and RL, good conservation practices were adopted, resulting in the most effective scenario in terms of reducing sediment generation and transport, reaching up to 65% reduction in transport in critical areas. This result highlights the need for adequate pasture management to control erosion processes.

In the context studied, the most vulnerable sub-basins were Caratinga, Piracicaba, Casca, Santo Antônio, and Carmo, which presented strong and/or very strong susceptibility to erosion and high production and transport of sediments, highlighting the urgent need for environmental restoration and recovery actions, with an emphasis on soil conservation. However, it is worth noting that the Suaçuí Grande sub-basin, despite its strong and very strong susceptibility, presented lower sediment production, possibly because it has lower precipitation values and is a region of lesser slope.

The effective implementation of the Forest Code (APP and RL) and the adoption of conservation practices in pasture areas are essential to mitigate the degradation of the basin. The results obtained and the methodology used can contribute to the application of reparation and mitigation measures in the Doce River basin. It also reinforces the importance of using hydrosedimentological modeling as a tool for environmental management and planning of river basins.

Our results effectively support recovery efforts within the framework of the New Rio Doce Agreement, identifying priority sub-basins where interventions have yielded the greatest benefits in terms of hydro-sedimentological aspects.

Acknowledgements The present study was carried out as part of the Aquatic Biodiversity Monitoring Program,-PMBA, established by the Technical-Scientific Agreement (001/2018) between UFES-FEST-Renova Foundation.

Author contributions MBR, ITS, LLR, SLC and MSM: Conceptualisation; Writing—original draft; Writing—revision; Methodology; Data curation; Formal analysis; Investigation.

DCB, ASFM, DR and MAC: Writing—review and editing; Supervision; Validation. All authors have read and agreed to the submitted version of the manuscript.

Funding Not applicable.

Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval All authors have read, understood, and have complied as applicable with the statement on "Ethical responsibilities of Authors" as found in the Instructions for Authors.

Competing interests The authors declare no competing interests.

References

Admas, B. F., Gashaw, T., Adem, A. A., Worqlul, A. W., Dile, Y. T., & Molla, E. (2022). Identification of soil erosion hot-spot areas for prioritization of conservation measures using the SWAT model in Ribb watershed, Ethiopia. *Resources, Environment and Sustainability, 8*, Article 100059. https://doi.org/10.1016/j.resenv.2022.100059

Aires, U. R. V., Silva, DDda, Fernandes Filho, E. I., Rodrigues, L. N., Uliana, E. M., Amorim, R. S. S., Ribeiro, CBdeM., & Campos, J. A. (2022). Modeling of surface sediment concentration in the Doce River basin using satellite remote sensing. *Journal of Environmental Management, 323*, Article 116207. https://doi.org/10.1016/j.jenvman.2022.116207

Aires, U. R. V., Silva, DDda, Fernandes Filho, E. I., Rodrigues, L. N., Uliana, E. M., Amorim, R. S. S., Ribeiro, CBdeM., & Campos, J. A. (2023). Machine learning-based modeling of surface sediment concentration in Doce River basin. *Journal of Hydrology, 619*, Article 129320. https://doi.org/10.1016/j.jhydrol.2023.129320

Alexandridis, T. K., Sotiropoulou, A. M., Bilas, G., Karapetsas, N., & Silleos, N. G. (2015). A comparative analysis of the R, K, LS, C and P factors in two soil erosion models (USLE and RUSLE) using GIS: A case study in Thessaly, Greece. *European Journal of Remote Sensing, 48*(1), 151–168. https://doi.org/10.5721/EuJRS20154809

Agência Nacional de Águas – ANA. (2024). *HidroWeb: Sistemas de informações hidrológicas*. Retrieved March 5, 2024, from http://hidroweb.ana.gov.br/HidroWeb

Andualem, T. G., Hewa, G. A., Myers, B. R., Peters, S., & Boland, J. (2023). Erosion and sediment transport modeling: A systematic review. *Land, 12*(7), 1396. https://doi.org/10.3390/land12071396

Anley, M. A., & Minale, A. S. (2024). Modeling the impact of land use land cover change on the estimation of soil loss and sediment export using InVEST model at the Rib watershed of Upper Blue Nile Basin, Ethiopia. *Remote Sensing Applications: Society and Environment, 34*, Article 101177. https://doi.org/10.1016/j.rsase.2024.101177

Arruda, V. L. S., Alencar, A. A. C., Carvalho Júnior, O. A., Ribeiro, F. F., Arruda, F. V., Conciani, D. E., Silva, W. V., & Shimbo, J. Z. (2024). Assessing four decades of fire behavior dynamics in the Cerrado biome (1985 to 2022). *Fire Ecology, 20*, Article 64. https://doi.org/10.1186/s42408-024-00298-4

Baeza, S., Vélez-Martin, E., Abelleyra, D., Banchero, S., Gallego, F., Schirmbeck, J., Veron, S., Vallejos, M., Weber, E., Oyarzabal, M., Barbieri, A., Petek, M., Lara, M. G., Sarrailhé, S. S., Baldi, G., Bagnato, C., Bruzzone, L., Ramos, S., & Hasenack, H. (2022). Two decades of land cover mapping in the Río de la Plata grassland region: The MapBiomas Pampa initiative. *Remote Sensing Applications: Society and Environment, 28*, Article 100834. https://doi.org/10.1016/j.rsase.2022.100834

Bayer, D. M., & Collischonn, W. (2013). Análise de sensibilidade do modelo MGB-IPH a mudanças de uso da terra. *Revista Brasileira de Recursos Hídricos, 18*, 1–12.

Bayer, M., Assis, P. C., Suizu, T. M., & Gomes, M. C. (2020). Mudança no uso e cobertura da terra na bacia hidrográfica do rio Araguaia e seus reflexos nos recursos hídricos: o trecho médio do rio Araguaia em Goiás. *Confins, 48*. https://doi.org/10.4000/confins.33972

Brasil. (2012). *Lei nº 12.651, de 25 de maio de 2012: Dispõe sobre a proteção da vegetação nativa*. Diário Oficial da União. Retrieved March 13, 2024, from https://www.planalto.gov.br/ccivil_03/_ato2011-2014/2012/lei/l12651.htm

Brasil. (2024). *SICAR – Sistema de Cadastro Ambiental Rural*. Retrieved April 9, 2024, from https://www.car.gov.br

Brito Neto, R. L., Tagliaferre, C., Lemos, O. L., Rocha, F. A., & Paula, Ade. (2021). Application of the MGB-IPH model in the Pardo River hydrographic basin in different scenarios of use and occupation of the soil. *Ciência Florestal, 31*(1), 191–213. https://doi.org/10.5902/1980509836095

Buarque, D. C. (2015). *Simulação da geração e do transporte de sedimentos em grandes bacias: Estudo de caso do rio Madeira* [Doctoral dissertation, Universidade Federal do Rio Grande do Sul]. Lume Repositório Digital. Retrieved August 14, 2023, from http://hdl.handle.net/10183/129875

Carmo, FFdo, Kamino, L. H. Y., Junior, R. T., Campos, ICde, Carmo, FFdo, Silvino, G., Castro, KJdaSXde., Mauro, M. L., Rodrigues, N. U. A., Miranda, MPdeS., & Pinto, C. E. F. (2017). Fundão tailings dam failures: The environment tragedy of the largest technological disaster of Brazilian mining in global context. *Perspectives in Ecology and Conservation, 15*(3), 145–151. https://doi.org/10.1016/j.pecon.2017.06.002

Chasia, S., Olang, L. O., Juma, B., & Sitoki, L. (2024). Understanding the linkages between land-use transitions and soil erosion/sediment deposition: A case study of the transboundary Sio-Malaba-Malakisi watershed in Kenya and Uganda. *Physics and Chemistry of the Earth, 133*, Article 103529. https://doi.org/10.1016/j.pce.2023.103529

Coe, M. T. (2000). Modeling terrestrial hydrological systems at the continental scale: Testing the accuracy of an atmospheric GCM. *Journal of Climate, 13*(4), 686–704. https://

doi.org/10.1175/1520-0442(2000)013<0686:MTHSAT>2.0.CO;2

Coelho, A. L. N. (2009). Bacia hidrográfica do Rio Doce (MG/ES): Uma análise socioambiental integrada. *Geografares, 7*, 141–146. https://doi.org/10.7147/GEO7.156

Collischonn, W., & Tucci, C. E. M. (2001). Simulação hidrológica de grandes bacias. *Revista Brasileira de Recursos Hídricos, 6*, 95–118. https://doi.org/10.21168/rbrh.v6n1.p95-118

Costa, R. C. A., Santos, R. M. B., Fernandes, L. F. S., de Melo, M. C., Valera, C. A., Valle Junior, RFdo, Silva, MMAPdeM., Pacheco, F. A. L., & Pissarra, T. C. T. (2023). Hydrologic response to land use and land cover change scenarios: An example from the Paraopeba River Basin based on the SWAT model. *Water, 15*(8), Article 1451. https://doi.org/10.3390/w15081451

Cunha, E. R., Santos, C. A. G., Silva, R. M., Panachuki, E., Oliveira, PTSde., Oliveira, NdeS., & Falcão, KdosS. (2022). Assessment of current and future land use/cover changes in soil erosion in the Rio da Prata basin (Brazil). *Science of the Total Environment, 818*, Article 151811. https://doi.org/10.1016/j.scitotenv.2021.151811

Dananto, M., Aga, A. O., Yohannes, P., & Shura, L. (2022). Assessing the water-resources potential and soil erosion hotspot areas for sustainable land management in the Gidabo Watershed, Rift Valley Lake Basin of Ethiopia. *Sustainability, 14*(9), 5262. https://doi.org/10.3390/su14095262

de Waal, J., Watson, A., Miller, J., & van Niekerk, A. (2024). Estimating impacts of land cover change on erosion in a data-scarce catchment: Bot River, South Africa. *Hydrological Sciences Journal*. https://doi.org/10.1080/02626667.2024.2398653

Didoné, E. J., Minella, J. P. G., & Piccilli, D. G. A. (2021). Reduced tillage systems as a conservation strategy to control water erosion in different slope positions. *Soil & Tillage Research, 213*, Article 105126. https://doi.org/10.1016/j.still.2021.105126

Eckhardt, K., Haverkamp, S., Fohrer, N., & Frede, H.-G. (2002). SWAT-G, a version of SWAT99.2 modified for application to low mountain range catchments. *Physics and Chemistry of the Earth, 27*(9–10), 641–644. https://doi.org/10.1016/S1474-7065(02)00048-7

ECOPLAN-LUME (Consórcio). (2010). Plano Integrado de Recursos Hídricos da Bacia do Rio Doce (PIRH). Relatório Final. Porto Alegre. Retrieved January 23, 2025, from https://cbhdoce.org.br/pirh-parh-pap/pirh

ENGECORPS ENGENHARIA S.A., (2021). Consolidação do estado da arte sobre a situação e a gestão de recursos hídricos na bacia. Relatório Técnico 1454-ANA-03-RH-RT-0001-R1. Disponível em: https://www.cbhdoce.org.br/wp-content/uploads/2022/03/1454-ANA-03-RH-RT-0001-R1_compressed.pdf. Accessed on: 26 mar. 2025.

ESA – European Space Agency. (2023). *The programme of european union copernicus*. Global and European Digital Elevation Model (COP-DEM). Retrieved December 15, 2023, from https://spacedata.copernicus.eu/pt/collections/copernicus-digital-elevation-model.

Fagundes, HdeO., Fan, F. M., & Paiva, R. C. D. (2019). Automatic calibration of a large-scale sediment model using suspended sediment concentration, water quality, and remote sensing data. *RBRH*. https://doi.org/10.1590/2318-0331.241920180127

Fagundes, H. O., Fan, F. M., Paiva, R. C. D., Siqueira, V. A., Buarque, D. C., Kornowski, L. W., Laipelt, L., & Collischonn, W. (2021). Sediment flows in South America supported by daily hydrologic-hydrodynamic modeling. *Water Resources Research*. https://doi.org/10.1029/2020WR027884

Fagundes, H. O., Paiva, R. C. D., Fan, F. M., Buarque, D. C., & Fassoni-Andrade, A. C. (2020). Sediment modeling of a large-scale basin supported by remote sensing and in-situ observations. *CATENA, 190*, Article 104535. https://doi.org/10.1016/j.catena.2020.104535

Fan, F. M., Buarque, D. C., Pontes, P. R. M., & Collischonn, W. (2015). Um mapa de unidades de resposta hidrológica para a América do Sul. In ABRH (Ed.), *XXI Simpósio Brasileiro de Recursos Hídricos* (pp. 1–8). Associação Brasileira de Recursos Hídricos. Retrieved September 15, 2023, from https://files.abrhidro.org.br/Eventos/Trabalhos/4/PAP019919.pdf

Fan, F. M., Collischonn, W., Sorribas, M. V., & Pontes, P. R. M. (2013). Sobre o início da rede de drenagem definida a partir dos modelos digitais de elevação. *Revista Brasileira de Recursos Hídricos, 18*(3), 241–257. https://doi.org/10.21168/rbrh.v18n3.p241-257

FAO – Food and Agriculture Organization of the United Nations. (1971) *Soil map of the world.* FAO. Retrieved October 19, 2023, from https://www.fao.org/4/as361e/as361e.pdf

Felippe, M. F., Magalhães Junior, A. P., Mendes, L. C., Carneiro, P. S., & Gontijo, B. M. (2016). Conexões geohistóricas e contemporâneas entre ocupação territorial, degradação ambiental e rarefação hídrica na Bacia do Rio Doce. *Revista Geografias, 12*(2), 203–222. https://doi.org/10.35699/2237-549X..13474

Ferreira Neto, L. C., Diniz, C. G., Maretto, R. V., Persello, C., Pinheiro, M. L. S., Castro, M. C., Sadeck, L. W. R., Fernandes Filho, A., Cansado, J., Souza, A. A. A., Feitosa, J. P., Santos, D. C., Adami, M., Souza-Filho, P. W. M., Stein, A., Biehl, A., & Klautau, A. (2024). Uncontrolled illegal mining and garimpo in the Brazilian Amazon. *Nature Communications, 15*, Article 9847. https://doi.org/10.1038/s41467-024-54220-2

FEST – Fundação Espírito Santense de Tecnologia, (2024a). Programa de monitoramento da biodiversidade aquática da área ambiental I – Porção Capixaba do Rio Doce e região marinha e costeira adjacente. Relatório Anual do PMBA/Fest – RA2023, RT-46/JUN 24. Vitória: Fest. Retrieved October 23, 2024, from https://drive.google.com/drive/folders/1B_MUdCfqfXH-K5xssSnb01NDr2tFMh5F.

FEST – Fundação Espírito Santense de Tecnologia. (2024b). Programa de monitoramento da biodiversidade aquática da área ambiental I – Porção Capixaba do Rio Doce e região marinha e costeira adjacente. Material Suplementar Ambiente Dulcícola – Tema: Integração flúvio-estuarina - Reflexos de ações na bacia do Doce nos sistemas aquáticos continental e marinho. In *Relatório Anual do PMBA/Fest.* Retrieved October 21, 2025, from https://fest.org.br/pmba_download_id=16022

Föeger, L. B., (2019). *Modelagem hidrossedimentológica de grandes bacias com propagação inercial de vazão: estudo de caso da bacia do rio Madeira* [Dissertação de mestrado, Universidade Federal do Espírito Santo]. Retrieved November 27, 2023, from https://sappg.ufes.br/tese_drupal//tese_13824_dissertacao_lbf_final.pdf

Föeger, L. B., Buarque, D. C., Pontes, P. R. M., Fagundes, HdeO., & Fan, F. M. (2022). Large-scale sediment modeling with inertial flow routing: Assessment of Madeira river basin. *Environmental Modelling & Software, 149*, Article 105332. https://doi.org/10.1016/j.envsoft.2022.105332

Gómez-Delgado, F., Roupsard, O., Le Maire, G., Taugourdeau, S., Pérez, A., van Oijen, M., Vaast, P., Rapidel, B., Harmand, J. M., Voltz, M., Bonnefond, J. M., Imbach, P., & Moussa, R. (2011). Modelling the hydrological behaviour of a coffee agroforestry basin in Costa Rica. *Hydrology and Earth System Sciences, 15*(1), 369–392. https://doi.org/10.5194/hess-15-369-2011

Hatje, V., Pedreira, R. M. A., de Rezende, C. E., Schettini, C. A. F., de Souza, G. C., Marin, D. C., & Hackspacher, P. C. (2017). The environmental impacts of one of the largest tailing dam failures worldwide. *Scientific Reports, 7*(1), 10706. https://doi.org/10.1038/s41598-017-11143-x

Huang, J., Wang, Q., Feng, S., Wang, B., Wen, J., & Zhang, G. (2024). Development of a distributed hydrological model combined with sediment transport calculations and its application to a medium-scale basin in northeast China. *Environmental Research Communications, 6*(10), Article 105009. https://doi.org/10.1088/2515-7620/ad84a6

IBGE – Instituto Brasileiro de Geografia e Estatística. (2006). *Mapa pedológico - Base de dados espacial*. Rio de Janeiro. Retrieved September 15, 2023, from https://www.ibge.gov.br/geociencias/informacoes-ambientais/pedologia/15829-solos.html

Jiang, K., Mo, S., Yu, K., Li, P., & Li, Z. (2024). Analysis on the relationship between runoff erosion power and sediment transport in the Fujiang River basin and its response to land use change. *Ecological Indicators, 159*, Article 111690. https://doi.org/10.1016/j.ecolind.2024.111690

Kitheka, J. U., Kitheka, L. M., & Njogu, I. N. (2022). Suspended sediment transport in a tropical river basin exhibiting combinations of land uses/land covers and hydroclimatic conditions: Case study of upper Athi Basin, Kenya. *Journal of Hydrology: Regional Studies, 41*, Article 101115. https://doi.org/10.1016/j.ejrh.2022.101115

Knoben, W. J. M., Freer, J. E., & Woods, R. A. (2019). Technical note: Inherent benchmark or not? Comparing Nash-Sutcliffe and Kling-Gupta efficiency scores. *Hydrology and Earth System Sciences, 23*, 4323–4331. https://doi.org/10.5194/hess-23-4323-2019

Kucharik, C. J., Foley, J. A., Délire, C., Fisher, V. A., Coe, M. T., Lenters, J. D., Young-Moiling, C., Ramankutty, N., Norman, J. M., & Gower, S. T. (2000). Testing the performance of a dynamic global ecosystem model: Water balance, carbon balance, and vegetation structure. *Global Biogeochemical Cycles, 14*(3), 795–825. https://doi.org/10.1029/1999GB001138

Kumar, S., & Mishra, A. (2015). Critical erosion area identification based on hydrological response unit level for effective sedimentation control in a river basin. *Water Resources Management, 29*(6), 1749–1765. https://doi.org/10.1007/s11269-014-0909-3

Kütter, V. T., Martins, G. S., Brandini, N., Cordeiro, R. C., Almeida, J. P. A., & Marques, E. D. (2023). Impacts of a tailings dam failure on water quality in the Doce River: The largest environmental disaster in Brazil. *Journal of Trace Elements and Minerals, 5*, Article 100084. https://doi.org/10.1016/j.jtemin.2023.100084

Lima, J. E. F. W., Lopes, W. T. A., Carvalho, N. de O., Vieira, M. R., & Silva, E. M. da. (2005). Suspended sediment fluxes in the large river basins of Brazil. In *Sediment Budgets 1: Proceedings of symposium S1 held during the Seventh IAHS Scientific Assembly at Foz do Iguaçu, Brazil, April 2005* (IAHS Publ. 291, pp. 355–364). International Association of Hydrological Sciences. Retrieved February 21, 2024, from https://iahs.info/uploads/dms/13042.48%20355-363%20S11-19%20Werneck%20et%20al.pdf

Liu, C., He, Y., Li, Z., Chen, J., & Li, Z. (2021). Key drivers of changes in the sediment loads of Chinese rivers discharging to the oceans. *International Journal of Sediment Research, 36*(6), 747–755. https://doi.org/10.1016/j.ijsrc.2020.05.005

Liu, J., Chen, L., Zhang, D., Pan, S., & Zhang, X. (2024). An annual sediment yield indicator based on precipitation–vegetation unmatched-degree developed in the Huangfu River basin. *Ecological Indicators, 166*, Article 112584. https://doi.org/10.1016/j.ecolind.2024.112584

Liu, Y., Wang, F., Lin, Y., Cao, L., Zhang, S., Ge, W., Han, J., Chen, H., & Shi, S. (2023). Assessing the contributions of human activities to runoff and sediment transport change: A method for break point identification in double mass curves based on model fitting. *Journal of Hydrology: Regional Studies, 50*, Article 101589. https://doi.org/10.1016/j.ejrh.2023.101589

Lopes, T. R., Nascimento, J. G., Pacheco, A. B., Duarte, S. N., Neale, C. M. U., & Folegatti, M. V. (2022). Estimation of sediment production and soil loss in a water supply basin for the metropolitan region of São Paulo – Brazil. *Journal of South American Earth Sciences, 118*, Article 103929. https://doi.org/10.1016/j.jsames.2022.103929

MapBiomas Project. (2024). *Collection 9 of the Annual Land Cover and Land Use Maps of Brazil (1985–2023)* . MapBiomas Data, V1. Retrieved July 11, 2024, from https://mapbiomas.org

Melo, M. M. A. P. (2014). *Modelagem da qualidade da água na bacia hidrográfica do Rio Piracicaba-MG*. Universidade Federal de Minas Gerais.

Mendonça, L. M., Blanco, C., & Cruz, Jd. S. (2025). Impact of climate change on rainfall erosivity in the Amazon and Cerrado biomes under CMIP6 scenarios. *Journal of Sedimentary Environments, 10*, 501–516. https://doi.org/10.1007/s43217-025-00237-6

Menezes, D., Minella, J. P. G., & Tassi, R. (2021). Monitoring sediment yield for soil and water conservation planning in rural catchments. *Environmental Monitoring and Assessment, 193*(11), Article 736. https://doi.org/10.1007/s10661-021-08901-w

Moriasi, D. N., Arnold, J. G., Van Liew, M. W., Bingner, R. L., Harmel, R. D., & Veith, T. L. (2007). Model evaluation

guidelines for systematic quantification of accuracy in watershed simulations. *Transactions of the ASABE, 50*(3), 885–900. https://doi.org/10.13031/2013.23153

Moriasi, D. N., Gitau, M. W., Pai, N., & Daggupati, P. (2015). Hydrologic and water quality models: Performance measures and evaluation criteria. *Transactions of the ASABE, 58*(6), 1763–1785. https://doi.org/10.13031/trans.58.10715

Mrad, D., Boukhari, S., Dairi, S., & Djebbar, Y. (2024). Modeling the sediment yield and estimating the best management practices in the Seybouse basin, Northeastern Algeria. *Water Science and Technology, 89*(6), 1497–1511. https://doi.org/10.2166/wst.2024.067

Neitsch, S. L., Arnold, J. G., Kiniry, J. R., & Williams, J. R. (2011). *Soil and Water Assessment Tool: Theoretical documentation version 2009*. Texas A&M University System, College of Agriculture and Life Sciences.

Nóbrega, M. T., Collischonn, W., Tucci, C. E. M., & Paz, A. R. (2011). Uncertainty in climate change impacts on water resources in the Rio Grande Basin, Brazil. *Hydrology and Earth System Sciences, 15*(2), 585–595. https://doi.org/10.5194/hess-15-585-2011

Oliveira, PAde, Blanco, C. J. C., Mesquita, A. L. A., Lopes, D. F., & Furtado Filho, M. D. C. (2021). Estimation of suspended sediment concentration in Guamá River in the Amazon region. *Environmental Monitoring and Assessment, 193*(1), Article 79. https://doi.org/10.1007/s10661-021-08901-w

Paiva, R. C. D., Collischonn, W., & Tucci, C. E. M. (2011). Large scale hydrologic and hydrodynamic modeling using limited data and a GIS based approach. *Journal of Hydrology, 406*(3–4), 170–181. https://doi.org/10.1016/j.jhydrol.2011.06.007

Panagos, P., Borrelli, P., Meusburger, K., Alewell, C., Lugato, E., & Montanarella, L. (2015). Estimating the soil erosion cover-management factor at the European scale. *Land Use Policy, 48*, 38–50. https://doi.org/10.1016/j.landusepol.2015.05.021

Pessoa, M. M. E. P., Kayser, R., & Collischonn, W. (2012). Integração do Modelo Hidrológico para Grandes Bacias MGB IPH e Sistemas de Informação Geográfica para suporte à decisão de outorga de direito de uso da água. *Anais de Evento Técnico ou Relatório Institucional*. https://doi.org/10.21168/rega.v9n2.p21-33

Pontes, P. R. M., Fan, F. M., Fleischmann, A. S., de Paiva, R. C. D., Buarque, D. C., Siqueira, V. A., Jardim, P. F., Sorribas, M. V., & Collischonn, W. (2017). MGB-IPH model for hydrological and hydraulic simulation of large floodplain river systems coupled with open source GIS. *Environmental Modelling & Software, 94*, 1–20. https://doi.org/10.1016/j.envsoft.2017.03.029

Qin, Y., Shi, P., Hu, Y., & Zhou, D. (2025). Landscape indices explain different responses of runoff and sediment yield to land use/cover change on the Loess Plateau, China. *CATENA, 249*, Article 108674. https://doi.org/10.1016/j.catena.2024.108674

Ragab, R., & Bromley, J. (2010). IHMS-integrated hydrological modelling system. Part 1. Hydrological processes and general structure. *Hydrological Processes, 24*(19), 2663–2680. https://doi.org/10.1002/hyp.7681

Rautela, K. S., Kumar, M., Khajuria, V., & Alam, M. A. (2021). Comparative geomorphometric approach to understand the hydrological behaviour and identification of the erosion prone areas of a coastal watershed using RS and GIS tools. *Research Square*. https://doi.org/10.21203/rs.3.rs-966853/v1

Reda, Y., Moges, A., & Kendie, H. (2024). Assessing the impacts of soil and water conservation practices on soil erosion and sediment yield using RUSLE and SWAT models in Northwestern Highlands of Ethiopia. *Environmental Challenges, 13*, Article 100818. https://doi.org/10.1016/j.envc.2023.100818

Rocha, W. J. S. F., Vasconcelos, R. N., Duverger, S. G., Costa, D. P., Santos, N. A., Rocha, R. O. F., Santana, M. M. M., Alencar, A. A. C., Arruda, V. L. S., Silva, W. V., Ferreira-Ferreira, J., Oliveira, M., Barbosa, L. S., & Cordeiro, C. L. (2024). Mapping burned area in the Caatinga biome: Employing deep learning techniques. *Fire, 7*(12), Article 437. https://doi.org/10.3390/fire7120437

Rodrigues, A. S. L., Malafaia, G., Costa, A. T., & Nalini Júnior, H. A. (2013). Background values for chemical elements in sediments of the Gualaxo do Norte River basin, MG, Brazil. *Revista de Ciências Ambientais, 7*(2), 15–32.

Silva, BFda, Rodrigues, RZdosS., Heiskanen, J., Abera, T. A., Gasparetto, S. C., Biase, A. G., Ballester, M. V. R., Moura, YMde, Piedade, SMdeS., Silva, AKdeO., & Camargo, PBde. (2023b). Evaluating the temporal patterns of land use and precipitation under desertification in the semi-arid region of Brazil. *Ecological Informatics, 77*, Article 102192. https://doi.org/10.1016/j.ecoinf.2023.102192

Silva, C. R.; Cunha, F. G. & Viglio, E. P. (2016). III Congresso da Sociedade de Análise de Risco Latino-Americana (SRA-LA) Rompimento da barragem de mineração de ferro em Mariana - Minas Gerais, Brasil: impactos fisico-quimicos. Retrieved November 14, 2024, from https://doi.org/10.1016/j.ecoinf.2023.102192

Silva, LSda, Ferraz, L. L., Sousa, LFde, Jesus, RMde, Santos, C. A. S., & Rocha, F. A. (2023a). Assessment of changes in land use and occupation on the hydrological regime of a basin in the west of Bahia. *Journal of South American Earth Sciences, 123*, Article 104218. https://doi.org/10.1016/j.jsames.2023.104218

Sime, C. H., & Abebe, W. T. (2022). Sediment yield modeling and mapping of the spatial distribution of soil erosion-prone areas. *Applied and Environmental Soil Science, 2022*, Article 4291699. https://doi.org/10.1155/2022/4291699

Siqueira, V. A., Fleischmann, A., Jardim, P. F., Fan, F. M., & Collischonn, W. (2016). IPH-Hydro tools: A GIS coupled tool for watershed topology acquisition in an open-source environment. *Revista Brasileira de Recursos Hídricos, 21*, 274–287. https://doi.org/10.21168/rbrh.v21n1.p274-287

Souza Júnior, C. M. S., Marengo, J., Ferreira, B., Ribeiro, J., Schirmbeck, L. W., Hirye, M., Cunha, A., Wiederhecker, H. C., & Latuf, M. O. (2024). Amazon severe drought in 2023 triggered surface water loss. *Environmental Research: Climate, 3*, Article 041002. https://doi.org/10.1088/2752-5295/ad7c71

Terassi, PMdeB., Oliveira-Júnior, JFde, Gois, Gde, Oscar Júnior, A. C., Sobral, B. S., Biffi, V. H. R., Blanco, C. J. C., Correia Filho, W. L. F., & Vijith, H. (2020). Rainfall

and erosivity in the municipality of Rio de Janeiro, Brazil. *Urban Climate, 33*, Article 100637. https://doi.org/10.1016/j.uclim.2020.100637

Towner, J., Cloke, H. L., Zsoter, E., Flamig, Z., Hoch, J. M., Bazo, J., Coughlan de Perez, E., & Stephens, E. M. (2019). Assessing the performance of global hydrological models for capturing peak river flows in the Amazon basin. *Hydrology and Earth System Sciences, 23*, 3057–3080. https://doi.org/10.5194/hess-23-3057-2019

Tsige, M. G., Malcherek, A., & Seleshi, Y. (2022). Evaluation of soil erosion using RUSLE and GIS in different land management scenarios in the Upper Awash Basin, Ethiopia. *International Soil and Water Conservation Research, 10*(3), 291–302. https://doi.org/10.1016/j.iswcr.2022.03.004

Wang, C., Fu, X., Zhang, X., Wang, X., Zhang, G., & Gong, Z. (2024). Modeling soil erosion dynamic processes along hillslopes with vegetation impact across different land uses on the Loess Plateau of China. *CATENA, 243*, Article 108202. https://doi.org/10.1016/j.catena.2024.108202

Watene, G., Kiplagat, L. K., Gathenya, J. M., & Chemelil, M. C. (2021). Estimating soil loss using MUSLE incorporating sediment delivery ratio and support practice factor under different land uses in the Upper Tana Basin, Kenya. *Sustainable Water Resources Management, 7*, Article 119. https://doi.org/10.1007/s40899-021-00632-1

Wei, Y., Jiao, J., & Li, Y. (2022). Evolution of historical sediment yield using check-dam systems as carriers: A case study in a restored agricultural catchment on the Loess Plateau, China. *CATENA, 210*, Article 105877. https://doi.org/10.1016/j.catena.2021.105877

Williams, J. R. (1975). Sediment-yield prediction with universal equation using runoff energy factor. *Present and Prospective Technology for Predicting Sediment Yield and Sources* (pp. 244–252). U.S. Department of Agriculture.

Williams, J. R. (1995). The EPIC Model. In V. P. Singh (Ed.), *Computer Models of Watershed Hydrology* (pp. 909–1000). Water Resources Pubns.

Yin, S., Gao, G., Ran, L., Lu, X., & Fu, B. (2021). Spatiotemporal variations of sediment discharge and in-reach sediment budget in the Yellow River from the headwater to the delta. *Water Resources Research, 57*(11), Article e2021WR030130. https://doi.org/10.1029/2021WR030130

Zhang, Q., Qin, W., Ding, L., Cao, W., Xu, H., Liu, Q., & Shi, C. (2023). Impact of land use/cover change (LUCC) on sediment connectivity in small watersheds based on a revised index algorithm. *Geoderma, 438*, Article 116622. https://doi.org/10.1016/j.geoderma.2023.116622

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.